

Four-derivative correction to the generic three-charge BMPV black hole at finite rotation

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ABSTRACT: We determine the first-order correction in the supersymmetric four-derivative coupling λ_3 to the generic three-charge BMPV black hole. We first recover the static solution in the harmonic/isotropic-like radius used by Ref. [1], then map it exactly to the areal-radius gauge used by the rotating solver. In that gauge we linearize the complete standard-Weyl curvature-squared invariant and integrate the electric equations at fixed physical charge. For three unequal charges, the solution is exact in the rotation parameter: the parity-even corrections terminate at j^2 , the parity-odd corrections terminate at j^3 , and direct substitution verifies all thirteen radial equations. An asymptotic surface-charge calculation fixes $Q_I = \pi\mu_I/(4G_5)$, $J_1 = J_2 = \pi j/(4G_5)$, with no correction to the relation between j and the physical angular momenta. The entropy on the regular branch is

$$S = \frac{\pi^2}{2G_5} \sqrt{\mu_1\mu_2\mu_3 - j^2} \left[1 + \frac{4\lambda_3(3\mu_1\mu_2\mu_3 - 4j^2)}{\mu_3(\mu_1\mu_2\mu_3 - j^2)} \right] + \mathcal{O}(\lambda_3^2).$$

We also compare the five-dimensional solution with the rotating heterotic solution. The corrected scalar map is proved symbolically for arbitrary unequal charges and rotation. A raw-first dualization gives a closed third vector, and a second generic symbolic certificate proves its complete potential-level target match after a target-determined closed shift, the final vector redefinition, and radial pullback. The older three-form-first nonclosure is retained as a negative diagnostic. The shift coefficients are recognized in the source profiles but have not been independently derived from a displayed universal off-shell redefinition. The parity-odd sector has also been re-audited retrospectively against the sealed version-1 implementation. Version 1 did not define its epsilon objects clearly in prose, but its code used the alternating density and the curved tensor with the required determinant factors. No five-dimensional result requires a sign reversal.

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1 Introduction and starting conventions

For gauge coupling g and rotation-branch sign $\varepsilon = \pm 1$, the three-charge Gutowski–Reall black hole [2, 3] takes the form

$$ds^2 = -f dt^2 - 2f^2 w dt \sigma_L^3 + \frac{dr^2}{f \mathfrak{g}} + \frac{r^2}{4} \left(\frac{1}{f} ((\sigma_L^1)^2 + (\sigma_L^2)^2) + f^2 h (\sigma_L^3)^2 \right), \quad (1.1)$$

$$A^I = f X^I dt + (U^I + f w X^I) \sigma_L^3, \quad (1.2)$$

$$X_I = \frac{f H_I}{3} \quad \text{or} \quad X^I = \frac{1}{f H_I}, \quad (1.3)$$

where

$$f = (H_1 H_2 H_3)^{-1/3}, \quad \mathfrak{g} = 1 + g^2 (r^2 + \alpha_1), \quad (1.4)$$

$$w = -\varepsilon \frac{g r^2}{2} \left(1 + \frac{\alpha_1}{r^2} + \frac{\alpha_2}{2r^4} \right), \quad h = \frac{\mathfrak{g}}{f^3} - \frac{4w^2}{r^2}, \quad (1.5)$$

with

$$H_I = 1 + \frac{\mu_I}{r^2}, \quad U^I = \varepsilon \frac{g}{2} (r^2 + \alpha_1 - \mu_I), \quad (1.6)$$

and

$$\alpha_1 = \mu_1 + \mu_2 + \mu_3, \quad \alpha_2 = \mu_1 \mu_2 + \mu_2 \mu_3 + \mu_3 \mu_1, \quad \alpha_3 = \mu_1 \mu_2 \mu_3. \quad (1.7)$$

The explicit $SU(2)$ one-forms used throughout are

$$\sigma_L^1 = \sin \psi d\theta - \cos \psi \sin \theta d\phi, \quad (1.8)$$

$$\sigma_L^2 = \cos \psi d\theta + \sin \psi \sin \theta d\phi, \quad (1.9)$$

$$\sigma_L^3 = d\psi + \cos \theta d\phi. \quad (1.10)$$

with $0 \leq \theta \leq \pi$, $0 \leq \phi < 2\pi$, and $0 \leq \psi < 4\pi$. The Maurer–Cartan relation needed below is

$$d\sigma_L^3 = -\sigma_L^1 \wedge \sigma_L^2. \quad (1.11)$$

For intermediate coordinate integrals, define the positive coordinate orientation of the three-sphere by

$$\sigma_L^1 \wedge \sigma_L^2 \wedge \sigma_L^3 = \sin \theta d\theta \wedge d\phi \wedge d\psi, \quad \int_{S_{\text{coord}}^3} \sigma_L^1 \wedge \sigma_L^2 \wedge \sigma_L^3 = 16\pi^2. \quad (1.12)$$

This is a coordinate convention, not the orientation induced on a physical horizon; the latter is fixed by the spacetime orientation below and is the opposite one. The round

metric $\frac{1}{4} \sum_i (\sigma_L^i)^2$ has unsigned volume $2\pi^2$. In terms of the two orthogonal rotation-plane angles,

$$\phi_1 = \frac{\psi + \phi}{2}, \quad \phi_2 = \frac{\psi - \phi}{2}. \quad (1.13)$$

The symbols ϕ_1, ϕ_2 denote angles. The two unconstrained STU scalars introduced below are denoted φ_1, φ_2 . Our $J_1 = J_2$ denotes the common angular momentum conjugate to these two angles. This convention is stated explicitly because some entropy papers instead use an $SU(2)$ linear combination or the sum of the two momenta.

The physical orientation of the flat four-dimensional base is

$$\text{vol}_4 = \frac{r^3}{8} \sin \theta \, dr \wedge d\theta \wedge d\psi \wedge d\phi = -\frac{r^3}{8} dr \wedge \sigma_L^1 \wedge \sigma_L^2 \wedge \sigma_L^3. \quad (1.14)$$

This is the orientation of Eqs. (3.7)–(3.8) of Ref. [4]; with it, $d[j\sigma_L^3/(2r^2)]$ is anti-self-dual. Therefore (r, θ, ϕ, ψ) , and hence $(t, r, \theta, \phi, \psi)$, is an orientation-reversing coordinate order. We nevertheless use the positive coordinate three-sphere measure $\int_{S^3_{\text{coord}}} \sigma_L^1 \wedge \sigma_L^2 \wedge \sigma_L^3 = 16\pi^2$ when displaying intermediate surface integrals. The physically induced orientation is the opposite one.

The spacetime metric has mostly-plus signature. Latin indices a, b, c, d, e label an oriented orthonormal frame. We normalize the flat-index Levi–Civita symbol by

$$\boxed{\epsilon_{abcde} \text{ (flat)} : \quad \epsilon_{01234} = +1, \quad \epsilon^{01234} = -1,} \quad a, b, c, d, e \in \{0, 1, 2, 3, 4\}. \quad (1.15)$$

The upper sign follows by raising the flat indices with $\eta_{ab} = \text{diag}(-1, 1, 1, 1, 1)$. For Greek curved indices, the Levi–Civita tensor is instead

$$\epsilon_{\mu\nu\rho\sigma\lambda} = e_\mu^a e_\nu^b e_\rho^c e_\sigma^d e_\lambda^e \epsilon_{abcde}, \quad \epsilon^{\mu\nu\rho\sigma\lambda} = g^{\mu\alpha} g^{\nu\beta} g^{\rho\gamma} g^{\sigma\delta} g^{\lambda\kappa} \epsilon_{\alpha\beta\gamma\delta\kappa}. \quad (1.16)$$

Thus the vielbein determinant, equivalently the factor $e = \sqrt{-g}$, belongs to the curved tensor. Because the coordinate order above reverses the physical orientation, its ordered curved components are

$$\epsilon_{tr\theta\phi\psi} = -e, \quad \epsilon^{tr\theta\phi\psi} = +\frac{1}{e}, \quad e = \sqrt{-g}. \quad (1.17)$$

This does not contradict Eq. (1.15): transforming to the oriented orthonormal frame has negative vielbein determinant and gives the stated flat-symbol components $(+1, -1)$. For parity-odd coordinate densities we use a lowercase epsilon with a tilde,

$$\tilde{\epsilon}^{tr\theta\phi\psi} = +1, \quad \tilde{\epsilon}^{\mu\nu\rho\sigma\lambda} = e \epsilon^{\mu\nu\rho\sigma\lambda}, \quad (1.18)$$

in this orientation-reversing chart. This $\tilde{\epsilon}^{\mu\nu\rho\sigma\lambda}$ is an alternating coordinate symbol (a density), not the curved tensor. Mathematica’s raw `Normal[LeviCivitaTensor[5]]` returns this numerical array. Dividing it by e gives the curved upper tensor. Keeping the flat symbol, curved tensor, chart orientation and alternating density distinct is essential for matching both the supergravity action and the positive gauge potential in Eq. (4.1) of Ref. [5].

With $F^I = dA^I$, the Gutowski–Reall fields above solve the Einstein equation

$$R_{\mu\nu} - \frac{1}{2}g_{\mu\nu}R = \sum_I \left[\frac{1}{2(X^I)^2} \left(F_{\mu\rho}^I F_{\nu}^{I\rho} - \frac{1}{4}g_{\mu\nu}(F_{\rho\sigma}^I)^2 \right) \right. \quad (1.19)$$

$$\left. + \frac{1}{2(X^I)^2} \left(\partial_\mu X^I \partial_\nu X^I - \frac{1}{2}g_{\mu\nu}(\partial_\rho X^I)^2 \right) + 2g_{\mu\nu} \frac{g^2}{X^I} \right]. \quad (1.20)$$

In the ungauged $g \rightarrow 0$ limit, the black hole solution reduces to

$$ds^2 = -f dt^2 + \frac{1}{f} \left(dr^2 + \frac{r^2}{4} ((\sigma_L^1)^2 + (\sigma_L^2)^2 + (\sigma_L^3)^2) \right), \quad (1.21)$$

$$A^I = f X^I dt, \quad X^I = \frac{1}{f H_I}, \quad (1.22)$$

with

$$H_I = 1 + \frac{\mu_I}{r^2}, \quad f = (H_1 H_2 H_3)^{-1/3}. \quad (1.23)$$

This is the standard spherically symmetric three-charge BPS black hole solution to the ungauged STU model.

The equations above reproduce the starting draft. For the rotating ungauged calculation below, we use the conventional BMPV warp normalization in which the temporal square is $-f^2(dt + \omega \sigma_L^3)^2$. This avoids confusing the original draft's shorthand $-f dt^2$ with the exponent variable used by the radial solver. The harmonic functions H_I , charges μ_I , one-forms σ_L^i , and rotation parameter j retain the notation of the starting file.

2 Two- and four-derivative STU theory

The bosonic fields are the metric $g_{\mu\nu}$, three gauge potentials A^I , and three constrained scalars X^I satisfying $X^1 X^2 X^3 = 1$. We use $e = \sqrt{-g}$ and

$$S = \frac{1}{16\pi G_5} \int d^5x e \left[\mathcal{L}_{2\partial} + \lambda_3 \mathcal{L}_{4\partial}^{(3)} \right] + \mathcal{O}(\lambda_3^2), \quad (2.1)$$

with

$$\mathcal{L}_{2\partial} = R - \frac{3}{2} a_{IJ}(X) \partial_\mu X^I \partial^\mu X^J - \frac{3}{4} a_{IJ}(X) F_{\mu\nu}^I F^{J\mu\nu} \quad (2.2)$$

$$+ \frac{1}{4} C_{IJK} \epsilon^{\mu\nu\rho\sigma\lambda} F_{\mu\nu}^I F_{\rho\sigma}^J A_\lambda^K, \quad (2.3)$$

where

$$a_{IJ} = 3X_I X_J - 2C_{IJK} X^K, \quad X_I = C_{IJK} X^J X^K. \quad (2.4)$$

The STU model has $C_{123} = 1/6$ and permutations; equivalently,

$$a_{IJ} = \frac{1}{3} \text{diag} \left((X^1)^{-2}, (X^2)^{-2}, (X^3)^{-2} \right). \quad (2.5)$$

Following the original STU paper, the two independent unconstrained scalars are defined by

$$X^1 = e^{-\varphi_1/\sqrt{6} - \varphi_2/\sqrt{2}}, \quad X^2 = e^{-\varphi_1/\sqrt{6} + \varphi_2/\sqrt{2}}, \quad X^3 = e^{2\varphi_1/\sqrt{6}}. \quad (2.6)$$

The active four-derivative term $\mathcal{L}_{4\partial}^{(3)}$ is one complete supersymmetric invariant in the Riemann field frame. It contains curvature-squared terms, the mixed $A^3 \wedge R \wedge R$ term, and the vector and scalar terms whose coefficients are fixed by supersymmetry. Every equation below follows from varying this complete invariant. We set $\lambda_1 = \lambda_2 = 0$ and use

$$\lambda_3 = \frac{\alpha'}{8} \quad (2.7)$$

when comparing with the heterotic normalization. The four-derivative completion is related to Refs. [6, 7], and the generic static solution used below was obtained in Ref. [1].

The Cassani-frame description contains the same physical deformation together with curvature-free vector-multiplet invariants. After the allowed field redefinitions, the rotating BPS solution depends only on the single coupling λ_3 . As a direct check, we evaluated the full Euler–Lagrange variation of the separate curvature-free invariant, including its F^4 terms, on the two-derivative BMPV background. All thirteen first-order source functions vanish identically. This does not apply to the vector terms inside $\mathcal{L}_{4\partial}^{(3)}$: they are required by the supersymmetric curvature-squared completion and remain in every equation.

The equality $\lambda_3 = \alpha'/8$ fixes the coupling convention. By itself it does not prove a field-by-field map from the rotating ten-dimensional string-frame solution to the five-dimensional Einstein/Riemann frame. That separate comparison is presented in Section 14.

2.1 Flat symbol, curved tensor and action density

The original large Mathematica archive uses the ordered (+1) alternating array in every parity-odd coordinate density. This is correct for the orientation-reversing chart fixed above. Its historical variable `epsSymbol` represents $\tilde{\epsilon}^{\mu\nu\rho\sigma\lambda} = e \epsilon^{\mu\nu\rho\sigma\lambda}$, where the epsilon on the right is the curved upper tensor. Consequently the displayed component term is the geometric supergravity term. In oriented differential-form language the complete two-derivative STU Chern–Simons sector is $-C_{IJK} A^K \wedge F^I \wedge F^J$, with all indices summed. More explicitly, the two symmetric terms with $(I, J, K) = (1, 2, 3)$ and $(2, 1, 3)$ give the oriented (K=3) slice $-A^3 \wedge F^1 \wedge F^2/3$ and the coordinate component of the scalar Lagrangian

$$\mathcal{L}_{\text{CS}}^{A^3 F^1 F^2} = \frac{1}{12} \epsilon^{\mu\nu\rho\sigma\lambda} F_{\mu\nu}^1 F_{\rho\sigma}^2 A_\lambda^3 = \frac{1}{12e} \tilde{\epsilon}^{\mu\nu\rho\sigma\lambda} F_{\mu\nu}^1 F_{\rho\sigma}^2 A_\lambda^3. \quad (2.8)$$

Its positive coordinate-array sign is precisely the negative oriented-form sign because this chart reverses the physical orientation. Combining the three cyclic representatives, which differ by total derivatives, gives the single representative $-A^3 \wedge F^1 \wedge F^2$; on the equal-charge truncation it becomes $-A \wedge F \wedge F$ directly. The equal-charge Maxwell equation is

$$d \star F + F \wedge F = 0, \quad (2.9)$$

and $A = H^{-1}(dt + j\sigma_L^3/(2r^2))$ solves it exactly in r and j with the base orientation (1.14).

Notebook 00A constructs an oriented vielbein with negative determinant, checks the flat-symbol components $\epsilon_{01234} = +1$ and $\epsilon^{01234} = -1$, derives the curved coordinate components above, and performs the exact Maxwell test. No gauge-field sign bridge is made:

the archived and physical vector potentials and their component corrections have identical signs. Notebooks 00 and 14 retain the compact alternating array in the coordinate calculation and now document its geometric meaning explicitly.

There is one historical qualification. The version-1 report wrote some parity-odd terms without first declaring whether its epsilon was the flat symbol, the curved tensor or the alternating density. We therefore audited the sealed version-1 notebooks rather than inferring a sign from that prose. Notebook 00 used the raw array directly only in an action density and used `epsSymbol/sdetg` for curved-tensor contractions. This includes the complete $W'_{IJKL}\epsilon F^I F^J F^K dX^L$ sector: its seven expanded coefficients and its overall plus sign agree with the source superinvariant. Notebook 14 used the corresponding `epsSymbol/(2 sdetg)` factor in the mixed-curvature surface charge. These are exactly the determinant factors required by Eqs. (1.16)–(1.18). Notebook 29 reproduces this object-type audit in a fresh kernel and concludes that version 1 was under-documented, but not sign-flipped. In particular, no version-1 metric, vector, Page-charge or angular-momentum result is reversed. The later entropy calculation is certified separately by notebooks 20 and 21 under the same convention.

3 Paper radius and areal proof gauge

The static solution of Ref. [1] is naturally written in the harmonic radial coordinate inherited from its multicentre construction. We denote that coordinate by ρ and call it *isotropic-like*. This word is deliberate: after the field redefinitions to the five-dimensional Einstein/Riemann frame, the corrected spatial metric in that paper is not strictly conformally flat. Its harmonic functions and logarithmic derivatives are

$$H_I(\rho) = 1 + \frac{\mu_I}{\rho^2}, \quad \ell_I(\rho) = \frac{d}{d\rho} \log H_I(\rho), \quad \ell = \ell_1 + \ell_2 + \ell_3. \quad (3.1)$$

In this coordinate the static metric is parameterized without abbreviation as

$$ds^2 = -e^{2\mathcal{F}_P} dt^2 + e^{2\mathcal{G}_P} d\rho^2 + e^{2\mathcal{H}_P} \frac{\rho^2}{4} \sum_{i=1}^3 (\sigma_L^i)^2, \quad (3.2)$$

where

$$\mathcal{F}_P = -\frac{1}{3} \log(H_1 H_2 H_3) + \lambda_3 \delta \mathcal{F}_P, \quad (3.3)$$

$$\mathcal{G}_P = \frac{1}{6} \log(H_1 H_2 H_3) + \lambda_3 \delta \mathcal{G}_P, \quad (3.4)$$

$$\mathcal{H}_P = \frac{1}{6} \log(H_1 H_2 H_3) + \lambda_3 \delta \mathcal{H}_P. \quad (3.5)$$

The radial and angular corrections are independent at first order. All formulas in this section are coefficients of $\lambda_3 = \alpha'/8$. Define

$$M_t = \frac{2}{9H_3} [7(\ell_1 - \ell_2)^2 - 2(\ell_1 + \ell_2)\ell_3 + 19\ell_3^2], \quad (3.6)$$

$$M_\Omega = \frac{4}{9H_3} [(\ell_1 - \ell_2)^2 + (\ell_1 + \ell_2)\ell_3 + 4\ell_3^2], \quad (3.7)$$

$$M_r = \frac{4}{9H_3} (\ell^2 + 3\ell\ell_3). \quad (3.8)$$

For the static logarithmic metric functions used later, the paper-coordinate corrections are

$$\begin{aligned} \delta\mathcal{F}_P &= -\frac{1}{2}M_t, \\ \delta\mathcal{G}_P &= \frac{1}{2}(M_\Omega - M_r), \\ \delta\mathcal{H}_P &= \frac{1}{2}M_\Omega. \end{aligned} \quad (3.9)$$

There is no static rotation one-form or Hopf-fibre squashing sector. After applying the vector-sign dictionary of the present STU conventions, the electric and scalar corrections are

$$\delta u_{1,P} = \frac{2\ell(\ell_1 + \ell_2)}{3H_1H_3}, \quad \delta u_{2,P} = \frac{2\ell(\ell_1 + \ell_2)}{3H_2H_3}, \quad \delta u_{3,P} = -\frac{2\ell\ell_3}{3H_3^2}, \quad (3.10)$$

$$\delta\varphi_{1,P} = -\frac{2\ell^2}{3\sqrt{6}H_3}, \quad \delta\varphi_{2,P} = 0. \quad (3.11)$$

Thus $A^I = (H_I^{-1} + \lambda_3 \delta u_{I,P})dt$ in the static sector. There is no off-diagonal angular one-form or Hopf-fibre squashing, so no angular correction variable is needed here.

The rotating solver uses a different radial gauge. Write its radius as r and make the first-order change of coordinates

$$\rho = r + \lambda_3 \xi(r), \quad \xi(r) = -\frac{\delta\mathcal{H}_P(r)}{\mathcal{H}'_0(r) + 1/r}, \quad \mathcal{H}_0 = \frac{1}{6} \log(H_1 H_2 H_3). \quad (3.12)$$

For any scalar-like ansatz function Y , and separately for the radial and angular metric functions, the transformed corrections are

$$\delta Y_A = \delta Y_P + \xi Y'_0, \quad (3.13)$$

$$\delta\mathcal{G}_A = \delta\mathcal{G}_P + \xi\mathcal{G}'_0 + \xi', \quad (3.14)$$

$$\delta\mathcal{H}_A = \delta\mathcal{H}_P + \xi(\mathcal{H}'_0 + 1/r) = 0. \quad (3.15)$$

Here P means the paper coordinate and A the areal proof gauge. The only nontrivial common denominator introduced by this map can be written as

$$\mathcal{D}(r) = \sum_{I < J} (\mu_I + r^2)(\mu_J + r^2) > 0 \quad (3.16)$$

on the physical exterior. Notebook 28 embeds Eqs. (3.9)–(3.15), proves the map for arbitrary unequal charges, and verifies every static field against the independent areal seed of notebook 13.

The inverse coordinate presentation is $r = \rho - \lambda_3 \xi(\rho)$. Applying that same, j -independent radial pullback to the complete rotating answer gives a coordinate system whose $j = 0$ limit is exactly the paper form and which remains exact in j at first order in λ_3 . We call it the *paper-matched rotating coordinate*. We do not call it a uniquely isotropic rotating gauge: no source paper supplies such a condition, and a further j -dependent radial diffeomorphism would remain possible. This coordinate freedom is independent of all Levi-Civita and parity-odd signs.

4 Rotating background and corrected ansatz

For the ungauged rotating solution define

$$H_I(r) = 1 + \frac{\mu_I}{r^2}, \quad f(r) = (H_1 H_2 H_3)^{-1/3}, \quad \omega_0(r) = \frac{j}{2r^2}, \quad \Theta_0 = dt + \omega_0 \sigma_L^3. \quad (4.1)$$

The two-derivative BMPV background [8] is

$$ds_0^2 = -f^2 \Theta_0^2 + f^{-1} \left[dr^2 + \frac{r^2}{4} \sum_{i=1}^3 (\sigma_L^i)^2 \right], \quad (4.2)$$

$$A_0^I = H_I^{-1} \Theta_0, \quad (4.3)$$

$$\varphi_1^{(0)} = \frac{1}{\sqrt{6}} \log \frac{H_1 H_2}{H_3^2}, \quad \varphi_2^{(0)} = \frac{1}{\sqrt{2}} \log \frac{H_1}{H_2}. \quad (4.4)$$

For the separate curvature-free invariant, the vanishing source can also be seen directly on this rotating background. Its rescaled field strengths obey

$$\tilde{F}^I - \tilde{F}^J = (H_1 H_2 H_3)^{-1/3} \Theta_0 \wedge d \log \frac{H_I}{H_J} = -e^0 \wedge d \log \frac{X_0^I}{X_0^J}. \quad (4.5)$$

Here $e^0 = f \Theta_0$ is the timelike orthonormal coframe member. The terms proportional to $d\Theta_0$ cancel. The invariant and the first variations that enter the field equations therefore vanish pointwise on BMPV, without expanding in j .

The corrected cohomogeneity-one ansatz is

$$ds^2 = -e^{2\mathcal{F}} (dt + \omega \sigma_L^3)^2 + e^{2\mathcal{G}} dr^2 + e^{2\mathcal{H}} \frac{r^2}{4} [(\sigma_L^1)^2 + (\sigma_L^2)^2 + e^{2\mathcal{K}} (\sigma_L^3)^2], \quad (4.6)$$

$$A^I = u_I(r) dt + v_I(r) \sigma_L^3. \quad (4.7)$$

The constrained scalars X^I are parametrized by $\varphi_1(r), \varphi_2(r)$ through Eq. (2.6). Each field is expanded to first order in λ_3 :

$$\mathcal{F} = \log f + \lambda_3 \delta \mathcal{F}, \quad \mathcal{G} = -\frac{1}{2} \log f + \lambda_3 \delta \mathcal{G}, \quad (4.8)$$

$$\mathcal{H} = -\frac{1}{2} \log f + \lambda_3 \delta \mathcal{H}, \quad \mathcal{K} = \lambda_3 \delta \mathcal{K}, \quad (4.9)$$

$$\omega = \omega_0 + \lambda_3 \delta \omega, \quad \varphi_1 = \varphi_1^{(0)} + \lambda_3 \delta \varphi_1, \quad (4.10)$$

$$\varphi_2 = \varphi_2^{(0)} + \lambda_3 \delta \varphi_2, \quad u_I = H_I^{-1} + \lambda_3 \delta u_I, \quad (4.11)$$

$$v_I = \omega_0 H_I^{-1} + \lambda_3 \delta v_I. \quad (4.12)$$

As announced in Section 3, the equation solver uses areal-radius gauge,

$$\delta \mathcal{H} = 0, \quad (4.13)$$

and impose no second radial gauge condition.

At two derivatives the gauge potentials are aligned with Θ_0 . At first order it is therefore useful to separate the part of the angular potential induced by the electric correction and by the corrected co-rotating one-form:

$$A^I = (H_I^{-1} + \lambda_3 \delta u_I) [dt + (\omega_0 + \lambda_3 \delta \omega) \sigma_L^3] + \lambda_3 \beta_I \sigma_L^3 + \mathcal{O}(\lambda_3^2). \quad (4.14)$$

Comparison with $A^I = u_I dt + v_I \sigma_L^3$ gives

$$\beta_I = \delta v_I - \omega_0 \delta u_I - H_I^{-1} \delta \omega. \quad (4.15)$$

Thus β_I is the genuine first-order Hopf-fibre gauge correction after the two induced pieces have been removed; it is not simply δv_I . Because $d\sigma_L^3 \neq 0$, even a constant β_I changes the field strength.

Rotation parity makes the corrections to $u_I, \varphi_1, \varphi_2, \mathcal{F}, \mathcal{G}, \mathcal{K}$ even in j , whereas β_I and $\delta \omega$ are odd. We use the uniform coefficient convention below, where $\mathcal{Y}$ denotes any one of these fields:

$$\delta \mathcal{Y}_{\text{even}}(r, j) = \delta \mathcal{Y}^{(0)}(r) + j^2 \delta \mathcal{Y}^{(2)}(r) + \dots, \quad (4.16)$$

$$\delta \mathcal{Y}_{\text{odd}}(r, j) = j \delta \mathcal{Y}^{(1)}(r) + j^3 \delta \mathcal{Y}^{(3)}(r) + \dots. \quad (4.17)$$

5 Linearized equations and boundary conditions

After imposing $\delta \mathcal{H} = 0$, the first-order equations are written for

$$\{\delta u_I, \beta_I, \delta \varphi_1, \delta \varphi_2, \delta \mathcal{F}, \delta \mathcal{G}, \delta \mathcal{K}, \delta \omega\}. \quad (5.1)$$

The original angular potentials are reconstructed at the end from Eq. (4.15).

Substitution in the Euler–Lagrange equations gives thirteen linear inhomogeneous radial equations. Their left-hand sides are the two-derivative STU equations linearized about the BMPV background. Their source terms are the Euler–Lagrange variation of the complete λ_3 invariant evaluated on that background. The sources are lengthy, but the linear differential operators are compact and are printed explicitly in the equation notebook.

Each electric equation is a total radial derivative. Integrating once introduces the three conserved electric charges. We solve for a deformation at fixed physical charges

$$Q_I = \frac{\pi\mu_I}{4G_5}; \quad (5.2)$$

hence the parameters μ_I are held fixed and no electric-charge integration mode is retained. We denote the dimensionless first-order constants returned by these three integrated equations by c_I^{Page} ; their normalization in terms of physical charges is given in Section 11. The three integrated relations determine $\delta u'_I(r)$. The radial Einstein constraint—the equation obtained by varying the radial metric component—is algebraic in $\delta\mathcal{G}$, so it is eliminated before the remaining differential equations are solved. The reduced system has nineteen first-order modes and its principal second-order block has rank eight.

The action determines the field equations. The rotating ansatz, radial gauge, fixed-charge condition, asymptotic flatness, and horizon regularity then select the black-hole solution. The earlier static STU notebooks solve the $j = 0$ sector of this same theory; they do not define a different Lagrangian.

6 Additional radial-constraint check

Before the Maxwell first integrals are imposed, the radial Einstein equation is linear in $\delta\mathcal{G}$. Its coefficient is nonzero for $r > 0$, $\mu_I > 0$, and $0 \leq j^2 < \mu_1\mu_2\mu_3$. Thus the algebraic elimination used in Section 5 is valid throughout the physical exterior. Independent substitution of the closed solution into the unreduced radial equation and the three integrated Maxwell equations gives zero residuals.

7 Static seed and exact equal-charge check

Equations (3.9)–(3.11) are the generic unequal-charge static solution of Ref. [1] in its harmonic/isotropic-like radius. Notebook 28 applies the exact radial diffeomorphism (3.12). The transformed functions $\{\delta u_I^{(0)}, \delta\varphi_1^{(0)}, \delta\varphi_2^{(0)}, \delta\mathcal{F}^{(0)}, \delta\mathcal{G}^{(0)}\}$ are the areal-gauge seed stored in notebook 11. Notebook 13 independently substitutes that seed into all thirteen static equations and obtains exact zero residuals for arbitrary μ_1, μ_2, μ_3 . Thus Professor Liu’s “isotropic-like versus Schwarzschild-like” observation is an exact coordinate statement, not a change of Lagrangian, solution branch, or Levi-Civita convention. This is the static starting point before any rotation coefficient is solved.

For $\mu_1 = \mu_2 = \mu_3 = q$, the full correction provides a short independent check that is

exact in rotation. The following functions are the coefficients of λ_3 :

$$\delta u_1 = \delta u_2 = -\frac{16(3j^2 + 2q^3 - 3q^2r^2)}{3(q + r^2)^4}, \quad (7.1)$$

$$\delta u_3 = -\frac{4(12j^2q + 8q^4 - 3j^2r^2 + 14q^3r^2 + 6q^2r^4)}{3(q + r^2)^5}, \quad (7.2)$$

$$\delta v_1 = \delta v_2 = \frac{40jq^2}{3(q + r^2)^4}, \quad (7.3)$$

$$\delta v_3 = \frac{2j(15j^2 + 11q^3 + 20q^2r^2 + 9qr^4)}{3(q + r^2)^5}, \quad (7.4)$$

$$\delta\varphi_1 = \frac{4\sqrt{6}(2j^2 - 3q^3 - 3q^2r^2)}{3(q + r^2)^4}, \quad \delta\varphi_2 = 0, \quad (7.5)$$

$$\delta\mathcal{F} = -\frac{4(12j^2q + 8q^4 + 18j^2r^2 + 13q^3r^2 + 5q^2r^4)}{3r^2(q + r^2)^4}, \quad (7.6)$$

$$\delta\mathcal{G} = \frac{16(3j^2q + 2q^4 + 12j^2r^2 + 5q^3r^2 + 3q^2r^4)}{3r^2(q + r^2)^4}, \quad (7.7)$$

$$\delta\mathcal{H} = 0, \quad \delta\mathcal{K} = 0, \quad (7.8)$$

$$\delta\omega = \frac{8j(3j^2 + 2q^3 + 11q^2r^2 + 9qr^4)}{3r^4(q + r^2)^3}. \quad (7.9)$$

The adapted angular gauge corrections are

$$\beta_1 = \beta_2 = -\frac{24jq}{(q + r^2)^3}, \quad \beta_3 = -\frac{18jq}{(q + r^2)^3}. \quad (7.10)$$

All thirteen equations vanish on substitution. The active single- λ_3 invariant distinguishes the third vector multiplet, so $\delta u_1 = \delta u_2 \neq \delta u_3$ and $\delta\varphi_1 \neq 0$ even though the background charges are equal.

8 Generic unequal-charge solution linear in rotation

We next solve the terms linear in j at arbitrary unequal charges. The same charge-dependent denominator recurs in every rotation coefficient, so we denote it by

$$\Delta(r) = 3r^4 + 2(\mu_1 + \mu_2 + \mu_3)r^2 + \mu_1\mu_2 + \mu_1\mu_3 + \mu_2\mu_3. \quad (8.1)$$

Define the three numerator polynomials

$$N_B = 36\mu_1\mu_2\mu_3 + r^2(14\mu_1\mu_2 + 29\mu_1\mu_3 + 29\mu_2\mu_3) + r^4(7\mu_1 + 7\mu_2 + 22\mu_3), \quad (8.2)$$

$$N_3 = 27\mu_1\mu_2\mu_3 + r^2(8\mu_1\mu_2 + 23\mu_1\mu_3 + 23\mu_2\mu_3) + r^4(4\mu_1 + 4\mu_2 + 19\mu_3), \quad (8.3)$$

$$N_\omega = 12\mu_1^2\mu_2^2\mu_3^2 \quad (8.4)$$

$$+ r^2(22\mu_1^2\mu_2^2\mu_3 + 40\mu_1^2\mu_2\mu_3^2 + 40\mu_1\mu_2^2\mu_3^2) \quad (8.5)$$

$$+ r^4(6\mu_1^2\mu_2^2 + 63\mu_1^2\mu_2\mu_3 + 63\mu_1\mu_2^2\mu_3 + 27\mu_1^2\mu_3^2 + 102\mu_1\mu_2\mu_3^2 + 27\mu_2^2\mu_3^2) \quad (8.6)$$

$$+ r^6(15\mu_1^2\mu_2 + 15\mu_1\mu_2^2 + 39\mu_1^2\mu_3 + 144\mu_1\mu_2\mu_3 + 39\mu_2^2\mu_3 + 60\mu_1\mu_3^2 + 60\mu_2\mu_3^2) \quad (8.7)$$

$$+ r^8(8\mu_1^2 + 26\mu_1\mu_2 + 8\mu_2^2 + 77\mu_1\mu_3 + 77\mu_2\mu_3 + 32\mu_3^2) + r^{10}(9\mu_1 + 9\mu_2 + 36\mu_3). \quad (8.8)$$

The fixed-charge solution with no r^{-2} correction to the asymptotic rotation one-form is

$$\beta_1^{(1)} = -\frac{2N_B}{3(r^2 + \mu_1)^2(r^2 + \mu_2)(r^2 + \mu_3)^2}, \quad (8.9)$$

$$\beta_2^{(1)} = -\frac{2N_B}{3(r^2 + \mu_1)(r^2 + \mu_2)^2(r^2 + \mu_3)^2}, \quad (8.10)$$

$$\beta_3^{(1)} = -\frac{2N_3}{3(r^2 + \mu_1)(r^2 + \mu_2)(r^2 + \mu_3)^3}, \quad (8.11)$$

$$\delta\omega^{(1)} = \frac{4N_\omega}{3r^4(r^2 + \mu_1)(r^2 + \mu_2)(r^2 + \mu_3)^2\Delta(r)}. \quad (8.12)$$

The original angular gauge-potential coefficients follow from Eq. (4.15):

$$\delta v_I^{(1)} = \beta_I^{(1)} + \frac{\delta u_I^{(0)}}{2r^2} + \frac{r^2\delta\omega^{(1)}}{r^2 + \mu_I}, \quad \delta v_I = j\delta v_I^{(1)} + \mathcal{O}(j^3). \quad (8.13)$$

The integration constants are fixed by asymptotic decay and fixed physical electric charges. The remaining homogeneous r^{-2} shift of the rotation one-form is set to zero here; Section 11 shows that this is exactly the choice of fixed physical angular momenta. Direct substitution in the four equations for $\{\beta_1^{(1)}, \beta_2^{(1)}, \beta_3^{(1)}, \delta\omega^{(1)}\}$ gives exact zero residuals. These fields form the rotation-odd radial subsystem, whose principal matrix has full rank for $r > 0$ and positive μ_I . The remaining nine equations have vanishing linear coefficient by rotation parity.

At the horizon,

$$\beta_1^{(1)}(0) = -\frac{24}{\mu_1\mu_3}, \quad \beta_2^{(1)}(0) = -\frac{24}{\mu_2\mu_3}, \quad \beta_3^{(1)}(0) = -\frac{18}{\mu_3^2}, \quad (8.14)$$

$$\lim_{r \rightarrow 0} r^4\delta\omega^{(1)} = \frac{16\mu_1\mu_2}{\mu_1\mu_2 + \mu_1\mu_3 + \mu_2\mu_3}. \quad (8.15)$$

At infinity,

$$\lim_{r \rightarrow \infty} r^6\beta_1^{(1)} = \lim_{r \rightarrow \infty} r^6\beta_2^{(1)} = -\frac{14\mu_1 + 14\mu_2 + 44\mu_3}{3}, \quad (8.16)$$

$$\lim_{r \rightarrow \infty} r^6\beta_3^{(1)} = -\frac{8\mu_1 + 8\mu_2 + 38\mu_3}{3}, \quad (8.17)$$

$$\lim_{r \rightarrow \infty} r^6\delta\omega^{(1)} = 4(\mu_1 + \mu_2 + 4\mu_3). \quad (8.18)$$

In particular, $\delta\omega$ has no r^{-2} tail at this order.

9 Quadratic rotation for generic unequal charges

The complete coefficient of λ_3 is expanded uniformly as

$$\delta u_I(r, j) = \delta u_I^{(0)}(r) + j^2\delta u_I^{(2)}(r), \quad \beta_I(r, j) = j\beta_I^{(1)}(r) + j^3\beta_I^{(3)}(r), \quad (9.1)$$

$$\delta\varphi_1(r, j) = \delta\varphi_1^{(0)}(r) + j^2\delta\varphi_1^{(2)}(r), \quad \delta\varphi_2(r, j) = \delta\varphi_2^{(0)}(r) + j^2\delta\varphi_2^{(2)}(r), \quad (9.2)$$

$$\delta\mathcal{F}(r, j) = \delta\mathcal{F}^{(0)}(r) + j^2\delta\mathcal{F}^{(2)}(r), \quad \delta\mathcal{G}(r, j) = \delta\mathcal{G}^{(0)}(r) + j^2\delta\mathcal{G}^{(2)}(r), \quad (9.3)$$

$$\delta\mathcal{K}(r, j) = j^2\delta\mathcal{K}^{(2)}(r), \quad \delta\omega(r, j) = j\delta\omega^{(1)}(r) + j^3\delta\omega^{(3)}(r). \quad (9.4)$$

The static coefficients were established in Section 7; the new rotation-dependent coefficients are short enough to display in full.

The three integrated electric equations determine $d\delta u_I^{(2)}/dr$. Their integration constants are fixed by asymptotic decay and fixed physical charges. The radial Einstein constraint determines $\delta\mathcal{G}^{(2)}$ algebraically. The squashing equation decouples:

$$\frac{d^2\delta\mathcal{K}^{(2)}}{dr^2} + \frac{3}{r}\frac{d\delta\mathcal{K}^{(2)}}{dr} - \frac{8}{r^2}\delta\mathcal{K}^{(2)} = 0, \quad \delta\mathcal{K}^{(2)} = c_+r^2 + \frac{c_-}{r^4}. \quad (9.5)$$

Asymptotic flatness removes c_+ , while horizon regularity removes c_- . Hence $\delta\mathcal{K}^{(2)} = 0$.

The quadratic scalar and metric coefficients are

$$\delta\varphi_1^{(2)} = \frac{8\sqrt{6} [3r^4 + \{5(\mu_1 + \mu_2) - 4\mu_3\}r^2 + 7\mu_1\mu_2 - 2(\mu_1 + \mu_2)\mu_3]}{3(r^2 + \mu_1)(r^2 + \mu_2)(r^2 + \mu_3)^2\Delta(r)}, \quad (9.6)$$

$$\delta\varphi_2^{(2)} = -\frac{24\sqrt{2}(\mu_2 - \mu_1)}{(r^2 + \mu_1)(r^2 + \mu_2)(r^2 + \mu_3)\Delta(r)}, \quad \delta\mathcal{K}^{(2)} = 0, \quad (9.7)$$

$$\delta\mathcal{F}^{(2)} = -\frac{8 [9r^6 + 8(\mu_1 + \mu_2 + \mu_3)r^4 + 7(\mu_1\mu_2 + \mu_1\mu_3 + \mu_2\mu_3)r^2 + 6\mu_1\mu_2\mu_3]}{r^2(r^2 + \mu_1)(r^2 + \mu_2)(r^2 + \mu_3)^2\Delta(r)}, \quad (9.8)$$

$$\delta\mathcal{G}^{(2)} = \frac{48}{r^2(r^2 + \mu_3)^2\Delta(r)^2} [\mu_3(\mu_1\mu_2 + \mu_1\mu_3 + \mu_2\mu_3) \quad (9.9)$$

$$+ 2\{\mu_1\mu_2 + 3\mu_3(\mu_1 + \mu_2) + 2\mu_3^2\}r^2 + 3\{2(\mu_1 + \mu_2) + 5\mu_3\}r^4 + 12r^6]. \quad (9.10)$$

The electric-potential coefficients are

$$\delta u_1^{(2)} = -\frac{16 [3\mu_1\mu_2\mu_3 + (4\mu_1\mu_2 + 4\mu_1\mu_3 + \mu_2\mu_3)r^2 + (5\mu_1 + 2\mu_2 + 2\mu_3)r^4 + 3r^6]}{(r^2 + \mu_1)^2(r^2 + \mu_2)(r^2 + \mu_3)^2\Delta(r)}, \quad (9.11)$$

$$\delta u_2^{(2)} = -\frac{16 [3\mu_1\mu_2\mu_3 + (4\mu_1\mu_2 + \mu_1\mu_3 + 4\mu_2\mu_3)r^2 + (2\mu_1 + 5\mu_2 + 2\mu_3)r^4 + 3r^6]}{(r^2 + \mu_1)(r^2 + \mu_2)^2(r^2 + \mu_3)^2\Delta(r)}, \quad (9.12)$$

$$\delta u_3^{(2)} = \frac{4 [-12\mu_1\mu_2\mu_3 + \{\mu_1\mu_2 - 11\mu_3(\mu_1 + \mu_2)\}r^2 + 2(\mu_1 + \mu_2 - 5\mu_3)r^4 + 3r^6]}{(r^2 + \mu_1)(r^2 + \mu_2)(r^2 + \mu_3)^3\Delta(r)}. \quad (9.13)$$

For arbitrary symbolic μ_I , direct substitution of these coefficients together with the static seed gives zero in all rotation-even equations and reproduces the j^2 terms of the equal-charge solution.

10 Cubic rotation and exact termination

The only coefficients not yet fixed in Eq. (9.4) are the cubic rotation-odd functions. Their four equations give

$$\boxed{\beta_1^{(3)} = \beta_2^{(3)} = \beta_3^{(3)} = 0, \quad \delta\omega^{(3)} = \frac{24}{r^4(r^2 + \mu_3)\Delta(r)}}. \quad (10.1)$$

For equal charges this becomes $\delta\omega^{(3)} = 8/[r^4(r^2 + q)^3]$, in agreement with Section 7.

To prove that no higher rotation coefficient is omitted, first note the largest power of j that occurs in each four-derivative source evaluated on the two-derivative BMPV background:

equation whose source is evaluated	highest power of j
three electric-potential equations and the φ_1 equation	j^4
the φ_2 equation	j^2
three angular gauge-potential equations	j^3
temporal- and radial-metric equations	j^6
rotation-one-form equation	j^7
areal-radius and squashing equations	j^8

Substitution of Eqs. (9.4) and (10.1) gives 9/9 zero residuals at each of j^4, j^6, j^8 in the rotation-even equations and 4/4 zero residuals at each of j^5, j^7 in the rotation-odd equations. Since neither the homogeneous operators nor the sources contain higher powers, the finite check is exhaustive. Therefore

$$\delta\mathcal{Y}(r, j) = \delta\mathcal{Y}^{(0)}(r) + j^2\delta\mathcal{Y}^{(2)}(r), \quad \mathcal{Y} \in \{u_I, \varphi_1, \varphi_2, \mathcal{F}, \mathcal{G}, \mathcal{K}\}, \quad (10.2)$$

$$\beta_I(r, j) = j\beta_I^{(1)}(r) + j^3\beta_I^{(3)}(r), \quad \delta\omega(r, j) = j\delta\omega^{(1)}(r) + j^3\delta\omega^{(3)}(r), \quad (10.3)$$

exactly in j at first order in λ_3 .

11 Physical electric charges and angular momenta

The physical electric charges are obtained from the Page integrals. With the action normalization $1/(16\pi G_5)$, the orientation of Eqs. (1.14)–(1.18), and the oriented-form gauge equation $d(a_{IJ}\star F^J) + C_{IJK}F^J \wedge F^K = 0$, the closed Page form is $a_{IJ}\star F^J + C_{IJK}A^J \wedge F^K$. Thus

$$\begin{aligned} Q_I &= \frac{3}{16\pi G_5} \int_{S_{\infty, \text{phys}}^3} (a_{IJ}\star F^J + C_{IJK}A^J \wedge F^K) \\ &= -\frac{3}{16\pi G_5} \int_{S_{\infty, \text{coord}}^3} (a_{IJ}\star F^J + C_{IJK}A^J \wedge F^K). \end{aligned} \quad (11.1)$$

with the corresponding supersymmetric four-derivative completion understood. The second line is the coordinate-measure convention used by the notebooks; the minus sign is solely the reversal between the coordinate and induced orientations. This uses the same $U(1)$ basis and the same positive gauge potentials as Eq. (4.1) of Ref. [5]. For the solution,

$$Q_I^{\text{phys}} = \frac{\pi\mu_I}{4G_5}, \quad \delta Q_I^{\text{phys}} = -\frac{\pi}{G_5}c_I^{\text{Page}}. \quad (11.2)$$

Thus $c_I^{\text{Page}} = 0$ means that the physical electric charges are held fixed.

One homogeneous solution is generated by an infinitesimal redefinition of the two-derivative rotation parameter, $j \mapsto j + \lambda_3 c_j$. Its only nonzero field corrections are

$$\delta\omega_{\text{hom}}(r) = \frac{c_j}{2r^2}, \quad \delta v_{I, \text{hom}}(r) = \frac{c_j}{2(r^2 + \mu_I)}. \quad (11.3)$$

It changes neither the electric charges nor any other field. The asymptotic normalization is

$$J_1 = J_2 = \frac{\pi j}{4G_5}. \quad (11.4)$$

If a particular solution has $\delta\omega_{\text{particular}} = a_J/r^2 + \mathcal{O}(r^{-4})$, fixed physical angular momentum requires

$$c_j = -2a_J = -2 \lim_{r \rightarrow \infty} r^2 \delta\omega_{\text{particular}}(r). \quad (11.5)$$

The complete generic four-derivative angular surface charge was evaluated in the regular asymptotic gauge $A_\psi^I \rightarrow 0$, with background subtraction, using the Wald–Tachikawa expression of Cassani–Ruipérez–Turetta. Its generalized Komar tensor contains both the covariant-curvature sector and the differentiated mixed term

$$L_{ARR} = \frac{1}{2} \epsilon^{\mu\nu\rho\sigma\alpha} A_\alpha R_{\mu\nu}{}^{\beta\gamma} R_{\rho\sigma\beta\gamma}, \quad (11.6)$$

with $R_{\mu\nu}{}^{\beta\gamma}$ carrying the first pair lowered and both double two-forms projected to the algebraic Riemann symmetry class. Let $k_{(4)}^{\mu\nu}[\xi_\psi]$ denote the first-order four-derivative contribution to the antisymmetric Wald–Tachikawa Noether potential for the rotational Killing vector $\xi_\psi = \partial_\psi$. Its tr component is the one integrated over a constant- t , constant- r three-sphere. Every potentially nondecaying asymptotic coefficient vanishes separately in the covariant and mixed sectors. The first nonzero radial density is

$$\frac{\sqrt{-g} k_{(4)}^{tr}[\xi_\psi]}{\sin \theta} = -\frac{2j(\mu_1 + \mu_2 + 4\mu_3)}{r^4} + \mathcal{O}(r^{-6}), \quad (11.7)$$

after the three-sphere measure convention is included. It therefore has zero boundary integral. The gauge and matter surface terms also vanish because

$$A_\psi = \mathcal{O}(r^{-2}), \quad F^2 = \mathcal{O}(r^{-6}), \quad R^2 = \mathcal{O}(r^{-8}). \quad (11.8)$$

The closed particular solution has no asymptotic mass, electric-charge, or angular-momentum tail, and the explicit four-derivative surface term vanishes at infinity. Consequently

$$\delta Q_I^{\text{phys}} = 0, \quad \delta J_1 = \delta J_2 = 0, \quad c_j = 0, \quad (11.9)$$

so the selected solution is expressed at fixed generic physical electric charges and fixed physical angular momenta.

12 Horizon regularity

The regular branch has

$$\mu_I > 0, \quad j^2 < \mu_1 \mu_2 \mu_3. \quad (12.1)$$

Although individual correction functions have poles at the unshifted coordinate origin, their residues combine into one displacement of the degenerate horizon. Introduce a centered radial coordinate by

$$r_{\text{old}}^2 = r^2 + \lambda_3 \rho_H, \quad \rho_H = \frac{16(3j^2 + 2\mu_1 \mu_2 \mu_3)}{\mu_3(\mu_1 \mu_2 + \mu_1 \mu_3 + \mu_2 \mu_3)}. \quad (12.2)$$

The temporal, radial and angular pole residues then cancel simultaneously.

Future-horizon ingoing and co-rotating coordinates may be constructed order by order in λ_3 . In those coordinates the metric, the two STU scalars, the regular-frame gauge potentials and all contractions $F_{\mu\nu}^I F^{J\mu\nu}$ are finite. Let ξ_ψ be the Hopf-fibre Killing vector in the regular co-rotating chart. For the horizon generator χ ,

$$\chi^2|_H = 0, \quad \chi \cdot \xi_\psi|_H = 0, \quad \partial_r \chi^2|_H = 0, \quad (12.3)$$

at orders λ_3^0 and λ_3^1 . The angular velocity and surface gravity receive no first-order shifts,

$$\Omega_H^{(1)} = 0, \quad \kappa^{(1)} = 0. \quad (12.4)$$

This regularity statement is perturbative at a fixed positive BMPV gap and is not uniform at the zero-area bound in Eq. (12.1).

13 Entropy

13.1 Direct result at fixed asymptotic charges

Let

$$P = \mu_1 \mu_2 \mu_3, \quad \Delta_H = P - j^2 > 0, \quad S_0 = \frac{\pi^2}{2G_5} \sqrt{\Delta_H}. \quad (13.1)$$

Evaluating the corrected horizon area, the covariant Wald term and the mixed gauge-gravitational Chern–Simons contribution gives

$$S = \frac{\pi^2}{2G_5} \sqrt{P - j^2} \left[1 + \frac{4\lambda_3(3P - 4j^2)}{\mu_3(P - j^2)} \right] + \mathcal{O}(\lambda_3^2). \quad (13.2)$$

This is the entropy at the fixed physical electric charges and angular momenta established in Section 11. The perturbative answer is not uniform at the zero-area limit $P = j^2$.

13.2 Decomposition of the direct result

All formulas below multiply the first-order coupling λ_3 . After applying the unique horizon-centering shift, the complete corrected metric gives

$$\frac{\delta S_{\text{area}}}{S_0 \lambda_3} = \frac{4(2j^4 - 7j^2 P + 4P^2)}{P \mu_3 \Delta_H}. \quad (13.3)$$

This includes both the shifted horizon location and the corrected induced metric; it is not an uncorrected-area evaluation.

The raw BMPV coframe is not adapted to the rotating horizon. If e^0, e^4 are its timelike and Hopf-fibre members, the normal/tangent split is obtained by

$$e^{\hat{0}} = \gamma(e^0 - u e^4), \quad e^{\hat{4}} = \gamma(e^4 - u e^0), \quad u = \frac{j}{\sqrt{P}}, \quad \gamma = \sqrt{\frac{P}{\Delta_H}}. \quad (13.4)$$

The horizon components then become

$$R_{\hat{0}\hat{1}\hat{0}\hat{1}}|_H = \frac{4P - 3j^2}{P^{4/3}}, \quad F_{\hat{0}\hat{1}}^I|_H = -\frac{2\sqrt{\Delta_H}}{P^{1/3}\mu_I}. \quad (13.5)$$

Using $X_H^I = P^{1/3}/\mu_I$ and $X_{I,H} = \mu_I/(3P^{1/3})$, direct evaluation of the complete covariant curvature and vector terms gives

$$\frac{\delta S_{\text{cov}}}{S_0 \lambda_3} = -\frac{4}{\mu_3}. \quad (13.6)$$

The mixed gauge–gravitational Chern–Simons sign requires one convention step that is easy to miss. For the Lorentz matrix trace,

$$\text{tr}(R \wedge R) = R^a{}_b \wedge R^b{}_a = -R^{ab} \wedge R_{ab}. \quad (13.7)$$

With the alternating coordinate density $\tilde{\epsilon}^{tr\theta\phi\psi} = +1$ in Eq. (1.18), the component term

$$\frac{\lambda_3}{2} \epsilon^{\mu\nu\rho\sigma\gamma} R_{\mu\nu ab} R_{\rho\sigma}{}^{ab} A_\gamma^3 = \frac{\lambda_3}{2e} \tilde{\epsilon}^{\mu\nu\rho\sigma\gamma} R_{\mu\nu ab} R_{\rho\sigma}{}^{ab} A_\gamma^3 \quad (13.8)$$

is $+2\lambda_3 A^3 \wedge \text{tr}(R \wedge R)$ as an oriented five-form. The apparent sign reversal follows because the displayed coordinate order reverses the physical orientation. Including the common $1/(16\pi G_5)$ action normalization, write Γ_N for the pullback of the horizon normal-bundle connection. The Tachikawa formula [9] is then

$$\begin{aligned} \delta S_{\text{CS}} &= +\frac{\lambda_3}{G_5} \int_{H_{\text{phys}}} \Gamma_N \wedge F^3 \\ &= -\frac{\lambda_3}{G_5} \int_{H_{\text{coord}}} \Gamma_N \wedge F^3. \end{aligned} \quad (13.9)$$

The second expression is the coordinate-measure implementation used in the notebooks. Explicitly,

$$\Gamma_N = \frac{j\sqrt{\Delta_H}}{2P} \sigma_L^3, \quad F^3|_H = \frac{j}{2\mu_3} d\sigma_L^3, \quad (13.10)$$

$$\int_{H_{\text{phys}}} \sigma_L^3 \wedge d\sigma_L^3 = +16\pi^2, \quad \int_{H_{\text{coord}}} \sigma_L^3 \wedge d\sigma_L^3 = -16\pi^2. \quad (13.11)$$

so

$$\frac{\delta S_{\text{CS}}}{S_0 \lambda_3} = \frac{8j^2}{P\mu_3}. \quad (13.12)$$

The physical Tachikawa sign and the induced-horizon orientation both differ from a calculation that treats (θ, ϕ, ψ) as physically oriented; the two changes cancel in the numerical entropy. No sign of A^3 is reversed.

Adding Eqs. (13.3), (13.6), and (13.12) gives

$$\boxed{S = \frac{\pi^2}{2G_5} \sqrt{P - j^2} \left[1 + \frac{4\lambda_3(3P - 4j^2)}{\mu_3(P - j^2)} \right] + \mathcal{O}(\lambda_3^2)}. \quad (13.13)$$

With $\alpha' = 8\lambda_3$, this has the same spin dependence as the asymptotic-charge expression derived in the next subsection. At $j = 0$ it reduces to $S/S_0 = 1 + 12\lambda_3/\mu_3$. The perturbative result is non-uniform at the classical zero-area bound $P = j^2$, which remains outside its domain. An independent coordinate reconstruction of the curvature, horizon frame, normal connection, area determinant and Chern–Simons sign reproduces the same answer.

13.3 Why published rotating predictions differ

With the direct result established, we can compare quantities called physical charge and angular momentum in different calculations. For the one-parameter equal-charge notation used in Ref. [10], define

$$\varpi = \frac{J}{Q^{3/2}}, \quad S_{\text{eq}}^{(0)} = 2\pi Q^{3/2} \sqrt{1 - \varpi^2}, \quad w_{\text{CY}} = \left(\frac{2}{9\kappa}\right)^{1/3}. \quad (13.14)$$

Here κ is the one-parameter Calabi–Yau intersection number, c_2 is the corresponding one-parameter second-Chern-class coefficient that controls the four-derivative expansion in those references, and w_{CY} is a charge-normalization constant; it is unrelated to the BMPV rotation one-form denoted by w in the starting draft. All four proposed first-order results can be written as

$$S_{\text{bh}} = S_{\text{eq}}^{(0)} \left[1 + \frac{3w_{\text{CY}}c_2}{16Q} g(\varpi) \right] + \mathcal{O}(c_2^2), \quad (13.15)$$

but they predict

$$g_{\text{Euler}}(\varpi) = 1 + \frac{\varpi^2}{3}, \quad g_{4\text{D}/5\text{D}}(\varpi) = \frac{1}{1 - \varpi^2}, \quad (13.16)$$

$$g_{\text{horizon}}(\varpi) = 1, \quad g_{\text{asym}}(\varpi) = \frac{1 - \frac{4}{3}\varpi^2}{1 - \varpi^2}. \quad (13.17)$$

The four expressions agree at $J = 0$, so neither the static curvature coefficient nor the static normalization explains the discrepancy.

The first result, Ref. [11], evaluated the direct Wald term from an Euler/curvature-squared interaction on the uncorrected geometry and explicitly left the corrected-area contribution unproved. It is therefore not a calculation of the complete supersymmetric invariant used here. Refs. [12, 13] used full supersymmetric near-horizon/off-shell methods, but their electric charges and angular momentum are obtained through four-dimensional reduction or horizon attractor/Noether equations. Ref. [14] instead Legendre transforms an asymptotically flat grand-canonical action with respect to asymptotic conserved quantities and obtains g_{asym} .

These are not automatically the same variables when the action contains ordinary and mixed gauge–gravitational Chern–Simons terms. Write the entropy at fixed solution parameters as $S(q, j) = S_{\text{eq}}^{(0)}(q, j) + \lambda_3 S_1(q, j) + \mathcal{O}(\lambda_3^2)$. If the corresponding physical charges obey

$$\begin{aligned} Q_{\text{phys}} &= Q_0(q, j) + \lambda_3 \Delta Q(q, j), \\ J_{\text{phys}} &= J_0(q, j) + \lambda_3 \Delta J(q, j), \end{aligned} \quad (13.18)$$

then the correction at fixed physical charges is

$$\delta S|_{Q_{\text{phys}}, J_{\text{phys}}} = S_1 - \partial_Q S_{\text{eq}}^{(0)} \Delta Q - \partial_J S_{\text{eq}}^{(0)} \Delta J. \quad (13.19)$$

Because $\partial_J S_{\text{eq}}^{(0)} \propto -J/\sqrt{Q^3 - J^2}$, an $\mathcal{O}(\lambda_3)$ shift in the definition of angular momentum generates precisely the nontrivial $(1 - \varpi^2)^{-1}$ dependence in Eq. (13.17). This also explains why the ambiguity disappears in the static limit.

Two independent developments favour the asymptotic-charge answer. The large-degree five-dimensional index of Ref. [10] selects g_{asym} , while the direct ten-dimensional heterotic computation of Ref. [4] finds, in its dimensionless charge convention,

$$\begin{aligned} S_{\text{het}} &= 2\pi \sqrt{Q_+ Q_- [Q_0 + 3 - \varpi^2 (Q_0 + 4)]}, \\ \varpi &= \frac{J_-}{\sqrt{Q_+ Q_- Q_0}}. \end{aligned} \tag{13.20}$$

Its controlled large-charge expansion has $g(\varpi) = g_{\text{asym}}(\varpi)$, in agreement with the Cassani Legendre-transform result. The unexpanded square root in Eq. (13.20) is numerically successful, but only its term linear in the four-derivative coupling is derived by the effective action. Moreover, the expansion is non-uniform as $\varpi^2 \rightarrow 1$.

For this package, fixed physical electric charges and fixed asymptotic angular momenta were established before the horizon calculation. Notebooks 20 and 21 then obtained g_{asym} directly from the five-dimensional Wald–Tachikawa entropy, and notebook 18 subsequently verifies the algebraic relations among the published conventions.

14 Exact heterotic solution and the rotating field-map test

This section compares the five-dimensional result with the heterotic solution of Ref. [4]. The comparison has two logically separate parts. Equation (3.31) of that reference is an exact-in-rotation ten-dimensional solution through first order in α' . Mapping it to the Einstein/Riemann variables of the STU action requires a sequence of reductions, dualizations and field redefinitions. We first record the literal three-form-first route, because its nonclosure is a useful diagnostic. We then use the order that is regular on the rotating solution space: dualize the exact raw quotient first, apply the closed solution-space map, and only then perform the final vector redefinition and radial pullback. This second route completes the target-matching third-vector identity for arbitrary unequal charges and arbitrary rotation through first order in α' . The required closed shift is determined from that comparison and recognized in the ten-dimensional source profiles; it has not been derived independently from a displayed universal field redefinition. A unique off-shell covariant extension to unrelated backgrounds is therefore not claimed.

14.1 Ten-dimensional input and parameter dictionary

In the notation of Ref. [4], the four radial functions needed for the reduction are

$$Z_0 = 1 + \frac{q_0}{r^2} + \alpha' \frac{q_0^2}{r^2(r^2 + q_0)^2}, \tag{14.1}$$

$$Z_+ = 1 + \frac{q_+}{r^2} - \frac{2\alpha'}{r^2(r^2 + q_0)(r^2 + q_-)} \left(q_+ q_- - \frac{2a^2}{r^2 + q_0} \right), \tag{14.2}$$

$$Z_- = 1 + \frac{q_-}{r^2}, \quad W = \frac{a}{2r^2} + \alpha' \frac{aq_0}{r^2(r^2 + q_0)^2}. \tag{14.3}$$

The printed “1+” before $a/(2r^2)$ in Eq. (3.31) of that paper is a typographical error: it violates asymptotic flatness and is absent from the paper’s own expansion. The expression

in Eq. (14.3) satisfies the two Bianchi equations and the remaining Einstein equation of that construction.

The two-derivative dictionary is fixed once and used throughout this section:

$$(\mu_1, \mu_2, \mu_3) = (q_+, q_-, q_0), \quad j = a, \quad \lambda_3 = \frac{\alpha'}{8}. \quad (14.4)$$

We retain a within the heterotic calculation and return to j only when comparing with the five-dimensional fields. The common radial coordinate is r ; no replacement by r^2 is made.

14.2 Frames and conventions

The field map is performed in the following physical order:

$$\begin{aligned} &\text{ten-dimensional heterotic string frame} \longrightarrow \text{raw five-dimensional quotient fields} \\ &\longrightarrow \text{corrected Kaluza–Klein variables} \longrightarrow \text{torsion-free five-dimensional variables} \\ &\longrightarrow \text{dualized variables} \longrightarrow \text{Einstein/Riemann-frame STU fields.} \end{aligned} \quad (14.5)$$

Here Σ denotes the Kaluza–Klein circle scalar and Φ the five-dimensional generalized heterotic dilaton. Neither is one of the canonically normalized STU scalars φ_1, φ_2 defined in Eq. (2.6).

The five-dimensional STU endpoint uses the same mostly-plus orientation as the rest of this report. With Latin indices denoting the oriented orthonormal frame, the flat symbol obeys

$$\epsilon_{01234} = +1, \quad \epsilon^{01234} = -1. \quad (14.6)$$

The intermediate reduction vectors below retain their natural heterotic signs. The curved lower tensor used in dualization is obtained from the flat symbol with five vielbeins, as in Eq. (1.16), and is distinct from the alternating coordinate density defined in Eq. (1.18). Because $(t, r, \theta, \phi, \psi)$ reverses the physical orientation, $\epsilon_{tr\theta\phi\psi} = -\sqrt{-g}$ while $\epsilon^{tr\theta\phi\psi} = \tilde{\epsilon}^{tr\theta\phi\psi} / \sqrt{-g}$. No overall gauge-field sign change is made when the physical STU vectors are identified.

We use the left-invariant one-forms σ_L^i everywhere and define

$$H_A(r) = 1 + \frac{q_A}{r^2} \quad (A = +, -, 0), \quad \omega_0(r) = \frac{a}{2r^2}, \quad \omega_0 = \omega_0 \sigma_L^3, \quad \Theta_0 = dt + \omega_0. \quad (14.7)$$

The corrected co-rotating one-form is $\Theta = dt + W \sigma_L^3$. Relative to Ref. [4], our mostly-plus convention uses

$$z_{\text{STU}} = -z_{\text{het}}, \quad H_{\text{STU}}^{(3)} = -H_{\text{het}}^{(3)}, \quad (14.8)$$

where $H^{(3)}$ is the heterotic three-form, not a harmonic function. With one Lorentz index raised and one lowered, the torsionful connections obey

$$\Omega_{+, \text{STU}}^a{}_b = -\omega_{-, \text{het}}^a{}_b, \quad R_{\text{STU}}(\Omega_+)^a{}_b = -R_{\text{het}}(\omega_-)^a{}_b. \quad (14.9)$$

Consequently curvature squares and $\text{Tr}(R \wedge R)$ keep their signs. In particular, α' is not reversed by the convention change.

The circle one-forms are

$$\eta_{\text{het}} = dz_{\text{het}} - \mathcal{A}, \quad \eta_{\text{STU}} = dz_{\text{STU}} + \mathcal{A} = -\eta_{\text{het}}. \quad (14.10)$$

This fixes the horizontal Lorentz–Chern–Simons decomposition to be

$$\omega_{3L,6}^{\text{hor}} = \omega_{3L,6}^{\text{base}} + \mathcal{A} \wedge \iota_{z_{\text{het}}} \omega_{3L,6}. \quad (14.11)$$

There are two signature-dependent Hodge operations in this chain. We label them before writing the leading horizontal three-form:

$$\begin{aligned} \mathcal{M}_0^{\text{het}} &:= \star_{4,\text{het}} dH_0 = -\frac{q_0}{4} \sigma_L^1 \wedge \sigma_L^2 \wedge \sigma_L^3, \\ \mathcal{M}_0^{\text{STU}} &:= \star_{4,\text{STU}} dH_0 = +\frac{q_0}{4} \sigma_L^1 \wedge \sigma_L^2 \wedge \sigma_L^3 = -\mathcal{M}_0^{\text{het}}. \end{aligned} \quad (14.12)$$

The leading horizontal three-form is therefore

$$\begin{aligned} h_{\text{STU}}^{(0)} &= -\mathcal{M}_0^{\text{het}} + (H_+ H_-)^{-1} d\omega_0 \wedge \Theta_0 \\ &= \mathcal{M}_0^{\text{STU}} + (H_+ H_-)^{-1} d\omega_0 \wedge \Theta_0. \end{aligned} \quad (14.13)$$

To compare this directly with the physical third STU vector, set $F^{3(0)} = d(H_0^{-1} \Theta_0)$. The five-dimensional Hodge operation in the convention fixed above gives the exact identity

$$e^{2\Phi_0} \star_5 F^{3(0)} = -\mathcal{M}_0^{\text{STU}} - (H_+ H_-)^{-1} d\omega_0 \wedge \Theta_0. \quad (14.14)$$

Thus Eq. (14.13) is precisely

$$h_{\text{STU}}^{(0)} = -e^{2\Phi_0} \star_5 F^{3(0)}. \quad (14.15)$$

The plus sign in the rotating term of $h_{\text{STU}}^{(0)}$ follows from the six-dimensional horizontal decomposition; the overall minus sign relative to the five-dimensional Hodge dual is fixed by the final dual-multiplier convention below.

14.3 Raw quotient fields

Write

$$Z_A = H_A + \alpha' z_A, \quad W = \omega_0 + \alpha' w_1. \quad (14.16)$$

The nonzero first-order profiles are

$$z_0 = \frac{q_0^2}{r^2(r^2 + q_0)^2}, \quad w_1 = \frac{aq_0}{r^2(r^2 + q_0)^2}, \quad (14.17)$$

$$z_+ = -\frac{2}{r^2(r^2 + q_0)(r^2 + q_-)} \left(q_+ q_- - \frac{2a^2}{r^2 + q_0} \right), \quad z_- = 0. \quad (14.18)$$

Completing the ten-dimensional Kaluza–Klein square gives the raw five-dimensional string-frame fields

$$ds_{5,s,\text{raw}}^2 = -(Z_+ Z_-)^{-1} \Theta^2 + Z_0 ds_{\mathbb{R}^4}^2, \quad (14.19)$$

$$\Sigma_{\text{raw}} = \frac{1}{2} \log \frac{Z_+}{Z_-}, \quad \Phi_{\text{raw}} = \frac{1}{4} \log \frac{Z_0^2}{Z_+ Z_-}, \quad (14.20)$$

$$\mathcal{A}_{\text{raw}} = Z_+^{-1} \Theta. \quad (14.21)$$

Equivalently, if $g_{5,s}^{(0)}$ is the leading string-frame quotient metric,

$$g_{5,s,\text{raw}} = \frac{Z_0}{H_0} \left[g_{5,s}^{(0)} + \frac{\Theta_0^2}{H_+ H_-} \right] - \frac{\Theta^2}{Z_+ H_-}. \quad (14.22)$$

The factor $1/H_0$ in the first term is fixed by the requirement $g_{5,s,\text{raw}}|_{\alpha'=0} = g_{5,s}^{(0)}$.

Two combinations recur in the expansion. To distinguish the full trace on the rotating background from its purely rotation-dependent part, define

$$\tau_{\text{full}} := \frac{z_+}{2H_+} = -\frac{q_+ q_- - 2a^2/(r^2 + q_0)}{(r^2 + q_+)(r^2 + q_-)(r^2 + q_0)}, \quad d_0 := \frac{z_0}{H_0} = \frac{q_0^2}{(r^2 + q_0)^3}. \quad (14.23)$$

They give

$$\Sigma_{\text{raw}} = \frac{1}{2} \log \frac{H_+}{H_-} + \alpha' \tau_{\text{full}}, \quad (14.24)$$

$$\Phi_{\text{raw}} = \frac{1}{4} \log \frac{H_0^2}{H_+ H_-} + \frac{\alpha'}{2} (d_0 - \tau_{\text{full}}), \quad (14.25)$$

$$\mathcal{A}_{\text{raw}} = H_+^{-1} \Theta_0 + \alpha' H_+^{-1} (w_1 \sigma_L^3 - 2\tau_{\text{full}} \Theta_0). \quad (14.26)$$

14.4 Corrected Kaluza–Klein variables

Let $\mathcal{A}$ and $\mathcal{B}$ be the momentum and winding vectors, with $\mathcal{F} = d\mathcal{A}$ and $\mathcal{G} = d\mathcal{B}$. Their leading values are

$$\mathcal{A}^{(0)} = H_+^{-1} \Theta_0, \quad \mathcal{B}^{(0)} = -H_-^{-1} \Theta_0. \quad (14.27)$$

The combinations entering the corrected circle reduction are

$$\mathcal{F}^\pm = e^\Sigma \mathcal{F} \pm e^{-\Sigma} \mathcal{G}, \quad T_{\text{KK}} = \frac{1}{32} (\mathcal{F}^-)^2 + \frac{1}{4} (\partial \Sigma)^2. \quad (14.28)$$

On the two-derivative rotating background,

$$\mathcal{F}^- = -(d \log H_+ + d \log H_-) \wedge e^0 + \frac{2}{\sqrt{H_+ H_-}} d\omega_0, \quad d\Sigma = \frac{1}{2} (d \log H_+ - d \log H_-), \quad (14.29)$$

where $e^0 = (H_+ H_-)^{-1/2} \Theta_0$ in this reduction frame. Since $(d\omega_0)_{\mathbb{R}^4}^2 = 16a^2/r^8$, one obtains

$$T_{\text{KK}} = -\frac{d \log H_+ \cdot d \log H_-}{4H_0} + \frac{(d\omega_0)_{\mathbb{R}^4}^2}{8H_+ H_- H_0^2} = \tau_{\text{full}}. \quad (14.30)$$

The corrected relation is $\Sigma_{\text{raw}} = \Sigma_{\text{KK}} + \alpha' T_{\text{KK}}$. Therefore

$$\Sigma_{\text{KK}} = \frac{1}{2} \log \frac{H_+}{H_-} + \mathcal{O}(\alpha'^2) \quad (14.31)$$

for arbitrary a . This is a symbolic, exact-in-rotation cancellation.

The corrected rotational profile also determines the only new duality type on the flat base. With

$$\text{vol}_4 = -\frac{r^3}{8} dr \wedge \sigma_L^1 \wedge \sigma_L^2 \wedge \sigma_L^3, \quad (14.32)$$

the signature bridge changes the Hodge star on odd-degree forms but not on two-forms. Hence the heterotic and STU stars agree in the following two-form identity:

$$\star_4(dr \wedge \sigma_L^3) = -\frac{r}{2} \sin \theta d\theta \wedge d\phi. \quad (14.33)$$

For $\omega = W(r)\sigma_L^3$,

$$d\omega + \star_4 d\omega = \left(W' + \frac{2W}{r}\right) \left[dr \wedge \sigma_L^3 - \frac{r}{2} \sin \theta d\theta \wedge d\phi\right], \quad (14.34)$$

and

$$w'_1 + \frac{2w_1}{r} = -\frac{4aq_0}{r(r^2 + q_0)^3}. \quad (14.35)$$

Thus the first-order rotation one-form contains one known selfdual two-form. It must be combined with the Lorentz–Chern–Simons contribution and the corrected winding-vector reduction; taken alone it is not a physical mismatch.

The vertical Lorentz–Chern–Simons calculation gives

$$[\text{Tr}(R_- \wedge R_-)]_{zr\theta\phi} = \frac{16aq_0r^3(q_0 + 3q_- + 4r^2) \sin \theta}{(q_0 + r^2)^4(q_- + r^2)^2}. \quad (14.36)$$

The contracted form satisfies $d(\iota_z \omega_{3L}) = \iota_z \text{Tr}(R_- \wedge R_-)$, and the convention bridge above fixes its sign relative to the corrected circle shift. The resulting winding vector is

$$\mathcal{B}_{\text{KK}} = -H_-^{-1} \Theta_0 + \alpha' \left(b_t^{(1)} dt + b_3^{(1)} \sigma_L^3\right) + \mathcal{O}(\alpha'^2), \quad (14.37)$$

$$b_t^{(1)} = -\frac{2r^2[-2a^2 + q_-q_+(q_0 + r^2)]}{(r^2 + q_-)^2(r^2 + q_+)(r^2 + q_0)^2}, \quad (14.38)$$

$$b_3^{(1)} = \frac{a[2a^2 - 2q_-q_+q_0 - (q_-q_+ + q_-q_0 + q_+q_0)r^2 - q_0r^4]}{(r^2 + q_-)^2(r^2 + q_+)(r^2 + q_0)^2}. \quad (14.39)$$

Both the quadratic and cubic powers of a are required. Differentiating Eq. (14.39) gives a closed, $SU(2)$ -invariant field strength with the correct static limit and no constant or $1/r^2$ tail.

14.5 Derivative-free and torsion-free variables

The next field redefinition removes derivatives of field strengths. If $h_{\mu\nu\rho}$ is the gauge-invariant horizontal three-form, the first-order shifts before multiplication by $\alpha'/8$ are

$$\mathcal{R}_{\mu\nu}^{(g)} = \mathcal{F}_{\mu\lambda}^- \mathcal{F}_\nu^{-\lambda} + \mathcal{F}_{\mu\lambda}^+ \mathcal{F}_\nu^{+\lambda} + 4\partial_\mu \Sigma \partial_\nu \Sigma, \quad (14.40)$$

$$\mathcal{R}^{(\Phi)} = \frac{1}{4} [(\mathcal{F}^-)^2 + (\mathcal{F}^+)^2] + (\partial\Sigma)^2, \quad (14.41)$$

$$\mathcal{R}_\mu^{(\mathcal{A})} = \frac{1}{2} e^{-\Sigma} \left(4\mathcal{F}_{\mu\nu}^+ \partial^\nu \Sigma + h_\mu^{\nu\lambda} \mathcal{F}_{\nu\lambda}^-\right), \quad (14.42)$$

$$\mathcal{R}_\mu^{(\mathcal{B})} = -\frac{1}{2} e^\Sigma \left(4\mathcal{F}_{\mu\nu}^- \partial^\nu \Sigma + h_\mu^{\nu\lambda} \mathcal{F}_{\nu\lambda}^+\right). \quad (14.43)$$

For example, $g_{\text{df}} = g_{\text{KK}} + \alpha' \mathcal{R}^{(g)}/8$, and similarly for the other fields.

The raw three-form contribution associated with this redefinition is

$$\mathcal{Q}_{\mu\nu\rho} := 3\mathcal{F}_{[\mu}^{+\lambda}\nabla_{\nu}^{\prime(+)}\mathcal{F}_{\rho]\lambda}^+. \quad (14.44)$$

Here $\nabla^{\prime(+)}$ is the predual covariant derivative with the torsionful connection $\Omega_+ = \omega + h/2$; the prime only distinguishes this intermediate frame from the final Riemann-frame derivative. Define

$$\mathcal{D}_{\text{map}}(r) = (q_- + r^2)^3(q_+ + r^2)^3(q_0 + r^2). \quad (14.45)$$

Its nonzero independent components in coordinates $(t, r, \theta, \phi, \psi)$ are

$$\mathcal{Q}_{tr\phi} = -\frac{2a(q_- - q_+)^2 r^5 \cos \theta}{\mathcal{D}_{\text{map}}(r)}, \quad \mathcal{Q}_{tr\psi} = -\frac{2a(q_- - q_+)^2 r^5}{\mathcal{D}_{\text{map}}(r)}, \quad (14.46)$$

$$\mathcal{Q}_{t\theta\phi} = -\frac{a(q_- - q_+)^2 r^6 \sin \theta}{\mathcal{D}_{\text{map}}(r)}, \quad \mathcal{Q}_{\theta\phi\psi} = -\frac{a^2(q_- - q_+)^2 r^4 \sin \theta}{2\mathcal{D}_{\text{map}}(r)}. \quad (14.47)$$

The remaining independent coordinate components vanish.

The torsion-free variables are then obtained in the same direction:

$$g_{\text{TF},\mu\nu} = g_{\text{df},\mu\nu} + \frac{\alpha'}{8}h_{\mu\rho\sigma}h_{\nu}^{\rho\sigma}, \quad \Phi_{\text{TF}} = \Phi_{\text{df}} + \frac{\alpha'}{32}h^2. \quad (14.48)$$

The sign and normalization are fixed by the symbolic static metric, generalized dilaton, momentum vector and three-form. A separate change of the lower-dimensional base two-form is

$$b_{\text{TF}} = b_{\text{KK}} - \frac{\alpha'}{8}\text{Tr}(\omega \wedge h). \quad (14.49)$$

This is distinct from the circle transgression already contained in the winding-vector reduction; applying both as the same shift would count one correction twice.

14.6 Dualization and the third vector

The third STU vector is introduced only after the torsion-free three-form has been assembled. We write $\tilde{h}$ for that torsion-free action variable, including the local lower-two-form and Lorentz–Chern–Simons shifts already described, but before eliminating it algebraically. Its corrected dualization equation is

$$\tilde{h}_{\alpha\beta\gamma} = -\frac{1}{2}e^{2\Phi_{\text{TF}}}\epsilon_{\mu\nu\alpha\beta\gamma}F^{3\mu\nu} + \frac{\alpha'}{8}\mathcal{J}_{\alpha\beta\gamma}, \quad \mathcal{J}_{\alpha\beta\gamma} := \frac{\delta\mathcal{L}_{4\partial}}{\delta h^{\alpha\beta\gamma}}. \quad (14.50)$$

The overall minus sign is not a gauge-potential redefinition. Here $\mathcal{C}$ is the one-form Lagrange multiplier, and after dualization it is the third STU potential with $F^3 = d\mathcal{C}$. The final source action uses

$$\Delta\mathcal{L} = +\mathcal{C} \wedge (dh + \mathcal{F} \wedge \mathcal{G}), \quad (F^1, F^2, F^3) = (\mathcal{F}, -\mathcal{G}, d\mathcal{C}). \quad (14.51)$$

Varying this action and using $\star_5^2 = -1$ on three-forms in Lorentzian signature gives the first term of Eq. (14.50). The same sign follows without components from

$$dh = F^1 \wedge F^2, \quad d(e^{2\Phi}\star_5 F^3) + F^1 \wedge F^2 = 0. \quad (14.52)$$

The source-file history contains both choices for the sign of the multiplier and of the field called F^3 . We therefore take the final action and the positive Eq. (4.1) potential as primary; copying the older plus-duality line while also using the final dictionary would mix the two conventions. The variational source $\mathcal{J}$ contains four separately evaluated parts: the quartic-three-form, curvature-three-form, quadratic-matter and linear-matter terms. Antisymmetric multiplicities are already present in both the kinetic contraction and the four-derivative scalar, so no additional factor of $3!$ is inserted in Eq. (14.50).

At leading order the torsion-free representative is $\tilde{h}^{(0)} = h_{\text{STU}}^{(0)}$. For an immediate component check, the static background gives

$$\left(e^{2\Phi_0} \star_5 F^{3(0)}\right)_{\theta\phi\psi} = -\frac{q_0}{4} \sin \theta, \quad \tilde{h}_{\theta\phi\psi}^{(0)} = +\frac{q_0}{4} \sin \theta. \quad (14.53)$$

Notebook 23 evaluates this relation with the curved lower Levi-Civita tensor and requires the minus branch to pass and the plus branch to fail.

As a static normalization test, take $q_- = 3$, $q_+ = 5$, $q_0 = 7$. The quartic-three-form and curvature-three-form contributions are

$$\mathcal{J}_{\theta\phi\psi}^{h^4} = -\frac{1715 \sin \theta}{(7+r^2)^3}, \quad \mathcal{J}_{\theta\phi\psi}^{Rh^2} = -\frac{147(7+2r^2) \sin \theta}{(7+r^2)^3}, \quad (14.54)$$

and the complete source is

$$\mathcal{J}_{\theta\phi\psi}^{\text{full}} = -\frac{98(28+3r^2) \sin \theta}{(7+r^2)^3}. \quad (14.55)$$

This static test fixes both the curvature contribution and the dualization normalization before any rotating comparison is made.

The static vector redefinitions on the two sides of the dualization are traversed in opposite directions. Their combined shift of the third-vector potential is

$$\delta \mathcal{C}_\mu^{\text{static}} = \frac{1}{3} F_{\mu\nu}^3 \partial^\nu \Phi, \quad (14.56)$$

where $\mathcal{C}$ denotes the potential that becomes A^3 . This identity, together with Eq. (14.50), proves the complete static map through the third vector.

14.7 Corrected generalized-dilaton map

The final scalar redefinition is most clearly tested after the dualization stage has been fixed. Define

$$\ell_\Sigma = \partial_r \log(H_+ H_-), \quad \ell_\Delta = \partial_r \log(H_+/H_-), \quad \ell_0 = \partial_r \log H_0. \quad (14.57)$$

For a generic static background, the torsion-free coefficient, first dualized shift, originally displayed final shift and the generalized dilaton in the published Riemann-frame solution

are

$$\Phi_{\text{TF}}^{(1)} = -\frac{\ell_\Sigma^2 + 2\ell_\Delta^2 - 10\ell_0^2}{32H_0}, \quad (14.58)$$

$$\delta_{\text{dual}}\Phi^{(1)} = \frac{\ell_0^2}{32H_0}, \quad (14.59)$$

$$\delta_{\text{R,old}}\Phi^{(1)} = \frac{8\ell_\Sigma^2 + 6\ell_\Delta^2 + 4\ell_\Sigma\ell_0 - 31\ell_0^2}{96H_0}, \quad (14.60)$$

$$\Phi_{\text{R}}^{(1)} = \frac{(\ell_\Sigma + \ell_0)^2}{48H_0}. \quad (14.61)$$

Their symbolic difference is

$$\Phi_{\text{TF}}^{(1)} + \delta_{\text{dual}}\Phi^{(1)} + \delta_{\text{R,old}}\Phi^{(1)} - \Phi_{\text{R}}^{(1)} = \frac{\ell_\Sigma^2}{32H_0}. \quad (14.62)$$

The leading field strength satisfies

$$e^{-8\Phi_0/3}(\mathcal{F}^{-(0)})^2 = -\frac{2\ell_\Sigma^2}{H_0}. \quad (14.63)$$

It follows that the correct final scalar redefinition is

$$\delta_{\text{R}}\Phi = \delta_{\text{R,old}}\Phi + \frac{1}{8}e^{-8\Phi/3}(\mathcal{F}^-)^2, \quad (14.64)$$

before multiplying the complete redefinition by $\alpha'/8$. Equivalently, the displayed combination

$$-\frac{3}{8}e^{-8\Phi/3}[(\mathcal{F}^+)^2 + (\mathcal{F}^-)^2] \quad (14.65)$$

is replaced by

$$-e^{-8\Phi/3} \left[\frac{3}{8}(\mathcal{F}^+)^2 + \frac{1}{4}(\mathcal{F}^-)^2 \right]. \quad (14.66)$$

The rotating statement can in fact be proved directly without substituting sample charges. Put

$$D_- = r^2 + q_-, \quad D_+ = r^2 + q_+, \quad D_0 = r^2 + q_0, \quad \mathcal{A}_q = (q_- + q_+)r^2 + 2q_-q_+. \quad (14.67)$$

Before the additional term in Eq. (14.64) is included, the complete mapped source minus the radially pulled-back five-dimensional target is

$$\mathcal{R}_\Phi = \frac{\mathcal{A}_q^2}{8D_-^2D_+^2D_0} - \frac{a^2}{D_-D_+D_0^2}. \quad (14.68)$$

A fresh symbolic contraction with the generic leading inverse metric gives

$$\frac{1}{64}e^{-8\Phi_0/3}(\mathcal{F}^{-(0)})^2 = -\mathcal{R}_\Phi. \quad (14.69)$$

Thus the corrected residual is identically zero for arbitrary (q_-, q_+, q_0, a) , wherever the displayed denominators are nonzero. Notebook 30 reconstructs the complete upstream

scalar chain, derives the radial pullback rather than assigning it, and proves Eqs. (14.68)–(14.69) in a fresh kernel. The five previous rational tuples are retained only as historical regression tests.

After all redefinitions, the Einstein-frame metric and constrained STU scalars are

$$g_E = e^{-4\Phi_R/3} g_{s,R}, \quad X^1 = e^{2\Phi_R/3-\Sigma}, \quad X^2 = e^{2\Phi_R/3+\Sigma}, \quad X^3 = e^{-4\Phi_R/3}. \quad (14.70)$$

When the dualization is integrable, the vector dictionary is

$$\boxed{(A^1, A^2, A^3) = (\mathcal{A}, -\mathcal{B}, \mathcal{C})}. \quad (14.71)$$

Here $\mathcal{C}$ is defined by the curved-lower-tensor, minus-sign duality relation in Eq. (14.50). This dictionary reproduces the positive STU potentials in Eq. (4.1) of Ref. [5]. The identity $X^1 X^2 X^3 = 1$ follows identically.

14.8 Raw-first completion of the rotating third vector

It is useful first to state why the earlier route was not used as the final map. If all predual three-form redefinitions are assembled literally and Eq. (14.50) is then inverted, the resulting two-form $\widehat{F}^3$ is not closed. For $(q_-, q_+, q_0, a, r) = (3, 5, 7, 1, 2)$, one component of $d\widehat{F}^3$, after removing common basis factors, is

$$\frac{46775306}{5752902771} \neq 0. \quad (14.72)$$

Tying every descendant of the same primitive profile gives coefficient and augmented ranks 12/13; fixing the primitive profiles to their physical normalizations gives 11/12. Hence a constant reweighting cannot turn this object into a field strength. Notebooks 23 and 27 retain this result as a negative diagnostic. In particular, an exact potential shift cannot “repair” a nonclosed two-form, because adding an exact field strength does not change its Bianchi identity.

The regular calculation instead begins by dualizing the exact raw quotient. For arbitrary radial functions Z_0, Z_+, Z_- and W , the horizontal three-form and raw quotient fields obey the algebraic identity

$$h_{\text{raw}} = -e^{2\Phi_{\text{raw}}} \star_{5,\text{phys}} d\mathcal{C}_{\text{raw}}, \quad \boxed{\mathcal{C}_{\text{raw}} = Z_0^{-1}(dt + W\sigma_L^3)}. \quad (14.73)$$

This follows directly by splitting the five-dimensional Hodge star into its timelike and four-dimensional-base pieces; it does not use a desired five-dimensional vector. Notebook 31 also checks all ten independent three-form components and the Bianchi identity through first order in α' . Expanding Eq. (14.73) gives

$$\mathcal{C}_{\text{raw}}^{(1)} = -\frac{q_0^2 r^2}{D_0^4} dt + \frac{aq_0(q_0 + 2r^2)}{2D_0^4} \sigma_L^3. \quad (14.74)$$

There are two independent stationary, $SU(2)$ -invariant potential components. Comparing the raw-first source with the independently equation-of-motion-certified five-dimensional target, while imposing the static anchor and a vanishing asymptotic gauge

tail, determines the following closed solution-space shift without any numerical charge substitution:

$$\Delta\mathcal{C}_{\text{map}} = T_C(r) dt + S_C(r) \sigma_L^3, \quad (14.75)$$

$$T_C(r) = \frac{a^2 r^2}{2D_- D_+ D_0^3}, \quad S_C(r) = \frac{a [a^2 - q_0 D_- D_+]}{4D_- D_+ D_0^3}. \quad (14.76)$$

The two functions are not interpolated from charge samples. They have the following exact decomposition in the primitive profiles of Eq. (14.3):

$$u_+ = H_+^{-1}, \quad u_0 = H_0^{-1}, \quad z_{+, \text{rot}}^{(1)} = \frac{4a^2}{r^2 D_0^2 D_-}, \quad w_1 = \frac{aq_0}{r^2 D_0^2}, \quad \omega_0 = \frac{a}{2r^2}, \quad (14.77)$$

$$T_C = \frac{1}{8} u_+ u_0 z_{+, \text{rot}}^{(1)}, \quad S_C = \omega_0 T_C - \frac{1}{4} u_0 w_1. \quad (14.78)$$

Equivalently, if $\tau_\omega = z_{+, \text{rot}}^{(1)}/(2H_+)$, then

$$\Delta\mathcal{C}_{\text{map}} = \frac{1}{4} [\tau_\omega \mathcal{C}^{(0)} - u_0 w_1 \sigma_L^3], \quad \mathcal{C}^{(0)} = u_0 \Theta_0. \quad (14.79)$$

Thus the target-determined shift admits an exact algebraic decomposition in the rotation-dependent momentum-harmonic and angular profiles of the ten-dimensional source; it is not an arbitrary radial fit. This recognition does not by itself derive the coefficients $1/8$ and $-1/4$ from a universal off-shell field-redefinition formula. The shift vanishes at $a = 0$, is even/odd in a in its temporal/fibre components, and falls as $T_C \sim a^2/(2r^8)$, $S_C \sim -aq_0/(4r^6)$. Since it is displayed at the potential level, $d\Delta\mathcal{C}_{\text{map}}$ is identically closed. In the invariant basis its fibre and sphere profiles obey $S'_C + \partial_r(-S_C) = 0$.

The remaining displayed vector redefinition is the already fixed postdual shift

$$\delta\mathcal{C}_\mu^{\text{post}} = \frac{1}{3} F_{\mu\nu}^{3(0)} \partial^\nu \Phi_0. \quad (14.80)$$

Let $\xi(r)$ be the generic radial coefficient derived in the coordinate map of notebook 28. The complete symbolic identity is

$$\boxed{\mathcal{C}_{\text{raw}}^{(1)} + \Delta\mathcal{C}_{\text{map}} + \delta\mathcal{C}^{\text{post}} = \mathcal{C}_{\text{STU,areal}}^{(1)} - \xi(r) \partial_r \mathcal{C}^{(0)}}. \quad (14.81)$$

Notebook 31 leaves (q_-, q_+, q_0, a, r) symbolic, embeds the independently certified target rather than substituting a numerical sample, and obtains zero for the temporal component, the Hopf-fibre component, and the complete five-component residual. Differentiating both sides proves the field-strength match and closure. Equation (14.81) therefore completes the rotating third-vector comparison on the generic BMPV solution space, exact in a through first order in α' .

The logical role of the older nonclosure is now precise. It shows that the literal three-form-first assembly is not a valid definition of the final rotating field strength; the raw-first route does not cancel that residual but avoids producing it. The new result is an exact

on-solution field map, not a derivation of Eqs. (14.75)–(14.76) from a unique manifestly covariant off-shell field redefinition valid on every five-dimensional background. In particular, the six constant structures $F_{\mu\nu}^I \nabla^\nu(\Phi, \Sigma)$ do not span this shift; a larger covariant basis may. A separate source-side test also rules out the only explicit discarded total-derivative current in Appendix B.4 of Ref. [15] that depends on the independent three-form. The source-formula ledger shows that the B.1–B.3 boundary currents use the plain torsion-free/composite derivative $\mathcal{D}$ and contain no independent h ; B.4 is the only displayed current among them with explicit h -dependence. Its induced canonical vector shift is proportional to $\star_5(\mathcal{F}^+ \wedge \mathcal{F}^+)$, but on the generic rotating source ansatz $\mathcal{F}^+ = dr \wedge v$, so every component of $\mathcal{F}^+ \wedge \mathcal{F}^+$ vanishes identically. Hence these displayed boundary terms do not derive $\Delta\mathcal{C}_{\text{map}}$. An implicit equation-of-motion field redefinition not printed in the reduction formulae could still account for it. The printed corrected KK scalar does not supply the missing derivation either: it is $T_{\text{KK}} = \tau_{\text{static}} + \tau_\omega$, whereas the temporal part of the required shift selects only $(\tau_\omega/4)\mathcal{C}^{(0)}$. Applying $(T_{\text{KK}}/4)\mathcal{C}^{(0)}$ universally would introduce a forbidden static term, and the displayed $-1/4$ correction to the winding vector is not a redefinition of the dual potential. Exact comparison also excludes absorbing the shift into either the radial pullback or the already displayed postdual vector redefinition. Finally, closure, rotation parity, the static anchor and falloff alone permit the two-parameter family

$$c_a u_0 \tau_\omega \Theta_0 + c_b u_0 w_1 \sigma_L^3, \quad (14.82)$$

and therefore do not select $c_a = 1/4$, $c_b = -1/4$. This proves underdetermination by the displayed and tested source-side structures rather than merely failing to find a formula within that set. Notebook 32 evaluates all of these no-go tests in a fresh kernel with generic charges and rotation. Deriving a universal extension would be useful for a general action-level dictionary, but it is separate from the exact field-by-field comparison of the two black-hole solutions.

Combining this result with the generic scalar proof above, the rotating heterotic-to-STU target identity is now symbolic for arbitrary unequal charges and exact in rotation at $\mathcal{O}(\alpha')$; the universal source-side origin of the closed third-vector shift remains underdetermined. Independently, the five-dimensional source equivalence of Section 2 continues to use $\lambda_1 = \lambda_2 = 0$ and $\lambda_3 = \alpha'/8$; the additional curvature-free heterotic vector invariants and their first variations vanish on the rotating BMPV background.

15 Conclusion

Writing $\mathcal{Y}$ for any field in the five-dimensional ansatz, the generic three-charge solution has the form

$$\mathcal{Y} = \mathcal{Y}_{\text{BMPV}}(j) + \lambda_3 \delta\mathcal{Y}_{\text{generic}}(r, j) + \mathcal{O}(\lambda_3^2), \quad (15.1)$$

where $\delta\mathcal{Y}_{\text{generic}}$ is symbolic in the three unequal charges and exact in j . The parity-even coefficients terminate at j^2 , the parity-odd coefficients at j^3 , and the finite source-degree test proves that no higher rotation coefficient is omitted. The solution is regular on the positive-gap branch and is written at fixed physical electric charges and fixed equal angular momenta. Its entropy is Eq. (13.2).

The static $j = 0$ answer is also given explicitly in the harmonic/isotropic-like coordinate of Ref. [1]. Notebook 28 proves its radial map to the areal gauge used by the equations. The same inverse pullback gives a paper-matched rotating presentation exact in j , while no claim is made that this presentation is the unique rotating isotropic gauge. The independent retrospective notebook-29 audit finds that version 1 was ambiguous only in its prose description of epsilon objects; its implemented parity-odd signs require no correction.

The five-dimensional equation-of-motion result is independent of the ten-to-five-dimensional comparison. In that comparison, the generic static and rotating scalar maps are proved symbolically. Raw-first dualization is proved independently of the target, and a closed target-determined solution-space shift makes the final third-vector residual vanish for arbitrary unequal charges and rotation. The remaining limitation is stated explicitly: the two shift coefficients have not been derived from a displayed universal off-shell field redefinition.

A Verification summary and notebook dictionary

The report uses physical notation throughout. Notebook 00A distinguishes the geometric Levi–Civita tensor from the alternating action density before the executable archive is used. The archive keeps the following shorter internal symbols:

notebook symbol	report quantity	meaning
x	r^2	notebook-only radial variable
q1,q2,q3	μ_1, μ_2, μ_3	harmonic-function parameters
jrot	j	rotation parameter
U1,U2,U3	$\delta u_1, \delta u_2, \delta u_3$	electric corrections
B1,B2,B3	$\beta_1, \beta_2, \beta_3$	co-rotating fibre corrections
Om	$\delta\omega$	rotation-one-form correction
Ft	$\delta\mathcal{F}$	temporal-warp correction
Gr	$\delta\mathcal{G}$	radial-warp correction
Sq	$\delta\mathcal{K}$	Hopf-fibre squashing correction
P	$-\sqrt{6}\delta\varphi_1/4$	legacy scalar coordinate
S	$\delta\varphi_2/\sqrt{2}$	legacy scalar coordinate

Claim	status	check
Two-derivative rotating background solves 13 equations	EXACT	13/13 zeros
Geometric Levi–Civita versus alternating action density	EXACT	notebook 00A; ordered components, metric raising, Eq. 4.1 Maxwell and mixed-CS checks

Claim	status	check
Retrospective version-1 parity-odd compatibility	EXACT	notebook 29; AFF , $W'\epsilon FFF dX$, and ARR signs and density factors, positive vector sign, and sealed-file provenance anchors
Homogeneous linearization has no field-independent remainder	EXACT	13/13 zeros
Electric equations are Page total derivatives	EXACT	3/3 identities
Page rules solve for $\delta u'_I$	EXACT	rank 3
Radial constraint eliminates $\delta\mathcal{G}$	EXACT	full Page-reduced identity and nonzero coefficient
Equal-charge solution solves original equations	EXACT	13/13 zeros
Generic static metric coefficient	EXACT	generic $j = 0$ 13/13 and exact gauge map
Paper isotropic-like radius to areal proof gauge	EXACT	notebook 28; every generic static field and inverse pullback, with $\lambda_3 = \alpha'/8$
Generic unequal-charge $\mathcal{O}(j)$ solution	EXACT	4 direct plus 9 parity
Generic unequal-charge $\mathcal{O}(j^2)$ solution	EXACT	9/9 raw equations
Generic unequal-charge $\mathcal{O}(j^3)$ solution	EXACT	4/4 odd zeros
Closure without new fields from j^4 through j^8	EXACT	35/35 residual zeros; 5 cancellations, 30 absence checks
Operator/source degree exhaustion	EXACT	no source above j^8
Equal-charge four-derivative angular falloff	EXACT	contained in generic Wald audit
Generic asymptotic Wald–Tachikawa angular charge	EXACT	covariant, mixed-CS, matter and Komar sectors
Generic invariant horizon regularity	EXACT	centered ingoing metric and finite invariants
Single- λ_3 Cassani matching	EXACT	independent standard-Weyl invariant; no cyclic sum
Heterotic extra-vector source cancellation	EXACT	invariant and first variation vanish, exact in j
Eq. (3.31) ten-dimensional profiles	EXACT	three ODE zeros and no asymptotic tail
Direct five-dimensional horizon entropy from the package fields	EXACT	notebooks 20 and 21; independent Wald–Tachikawa reconstruction

Claim	status	check
Heterotic entropy expansion gives g_{asym}	EXACT	subsequent algebra check in notebook 18
Raw rotating KK map and $T_{\text{KK}} = T_{\text{full}}$	EXACT	10/10 proved checks in notebook 19
Vertical winding/circle Lorentz-CS sector	EXACT	revised 19/19 ledger in notebook 22, exact in a
Horizontal convention, raw $\mathcal{Q}$, and $\Delta_{1,2}$ direction	EXACT	32/32 predual/convention checks in notebook 24
Independent lower-two-form representative	EXACT	notebook 25; exact in a , static direction fixed
Corrected second postdual scalar map	EXACT	notebooks 26 and 30; corrected coefficient, $\lambda_3 = \alpha'/8$, and generic rotating contraction
Corrected rotating scalar	EXACT	notebook 30; arbitrary unequal charges and rotation
Complete static field-map chain through the third vector	EXACT	corrected scalar coefficient plus static A^3 anchor
Final-action duality with positive A^3	EXACT	notebook 23: minus branch passes, plus branch fails
Algebraic corrected-dualization source	EXACT	notebook 23; four sectors, componentwise sum, fresh kernel
Literal combined rotating three-form/ A^3 chain	REJECTED ROUTE	notebook 27; exact nonzero closure residual
Tied-profile constant-sector repair	EXCLUDED	exact ranks 12/13, physical ranks 11/12
Raw-first rotating duality identity	EXACT	notebook 31; target-independent, arbitrary radial profiles
Closed rotating third-vector target match	EXACT IDENTITY	notebook 31; target-determined shift and zero generic potential residual
Final solution-level component/radial-gauge match	EXACT	notebook 31; all five components with symbolic charges and rotation
Independent source-side derivation of the two shift coefficients	NOT ESTABLISHED	notebook 32; full-KK-scalar rescaling, B.4 current, coordinate and displayed-postdual directions excluded

Claim	status	check
Unique off-shell covariant extension of the new map	NOT CLAIMED	separate action-level question outside the BMPV solution comparison

A.1 Staged reproduction path

The report presents the physics chronologically, with the generic static solution before rotation; the historical notebook numbers instead record a staged verification/dependency order. Run 00A (geometric Levi–Civita) and 29 (retrospective version-1 parity audit), then 00–01 (ansatz, action and equations), 02–03 (Maxwell reduction, constraint and source split), 04–05 (equal charges), 06–07 (generic linear rotation and physical normalization), 08–09 (quadratic and cubic sectors), and 10–12 (source-degree exhaustion, assembly and exact termination in j). Notebooks 13 and 28 are independent, after-the-fact certificates of the static result and its paper-coordinate bridge introduced earlier. Continue with 14–17 (normalization, horizon, source and Page checks), 20–21 (direct entropy), 18 (published-convention comparison), and finally 19 followed by 22–27 (historical field-map chain), 30 (generic rotating scalar proof), 31 (raw-first third-vector target identity), and 32 (source-origin no-go audit). The README specifies every notebook’s purpose, expected output and dependencies in this staged order.

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